

Operating-Boundary Routing Accountability and Transition-Window Reserve Diagnostics for Wind-Power Transitions

Junyu Li^a Juntao Du^{a,*}

^a School of Statistics and Applied Mathematics, Anhui University of Finance and Economics,
Bengbu, 233030, China

*Corresponding author.

Email addresses: ljylikezmn999@gmail.com (Junyu Li), dujuntao@aufe.edu.cn (Juntao Du)

Abstract

Wind-power forecasts matter most when they change reserve and shortage risk, not when they only lower an average error score. This paper studies one such operating interface: the maximum power point tracking (MPPT)-to-pitch transition, where the turbine response law changes and over-forecast errors can turn into reserve violations. We build a directed-diffusion gated recurrent unit (GRU) encoder with a node-level mixture-of-experts (MoE) gate, and we constrain the gate with load balancing, regime-anchor alignment, graph smoothness, wake supervision, and boundary-focused forcing. The KDD Cup 2022 wind-farm supervisory control and data acquisition (SCADA) benchmark (WTB) is the main control-boundary test; ERA5 reanalysis is used only as an observability contrast. The results support a bounded operating-diagnostic claim. Graph WaveNet remains the WTB accuracy reference, with overall root mean squared error (RMSE) 225.74 $\pm$ 2.60 and switch-window RMSE 228.12 $\pm$ 3.23, while the boundary-forced router pays a 10.39 overall-RMSE penalty. That penalty buys gate accountability: the router recovers the WTB operating partition with gate-regime normalized mutual information/adjusted Rand index (NMI/ARI) 0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371, and boundary-anchor removal collapses NMI/ARI by 0.7017 / 0.8593. Under a frozen validation-to-test reserve diagnostic at shortage-to-reserve cost ratio 10, the gate-bin policy is not a full-sample reserve policy, but on MPPT-to-pitch boundary anchors it lowers same-model total cost from 88.13M to 84.58M normalized reserve-energy cost units, violation rate from 0.1038 to 0.0900, and shortage energy from 3.73M to 3.19M, at the price of 1.85M additional reserve energy. The diagnostic also exposes failure: high-ramp windows can carry too little reserve, and Kelmarsh/Penmanshiel external-site runs fail the pre-specified routing criterion after local recalibration. A WTB proxy drill shows a usable workflow—small calibration window, local boundary and gate-map selection, and frozen held-out testing—inside the source benchmark only. The contribution is WTB internal boundary auditability, external failure-boundary diagnosis, and a transition-window reserve diagnostic.

Keywords: Wind-power operations; operating boundary; reserve decision; mixture of experts; regime routing; spatio-temporal graph learning.

1 Introduction

High wind penetration turns short forecast errors into reserve and dispatch exposure. System operators do not buy accuracy for its own sake; they buy enough flexible reserve to cover renewable uncertainty without paying avoidable reserve, shortage, imbalance, or curtailment costs [Doherty and O'Malley, 2005, Ela et al., 2011, Pinson, 2013]. Forecast value is therefore measured at the point where a prediction changes an operating decision: unit commitment, reserve procurement, market balancing, trading, or reliability screening [Pinson et al., 2007, Wang et al., 2011, Zhou et al., 2013]. In this paper, value is operationalized more narrowly as the cost-risk tradeoff induced by a frozen validation-to-test reserve quantile under transition-window shortfall. The setting is the maximum power point tracking (MPPT)-to-pitch boundary, where the turbine's local power-response law changes.

Average error is a weak first lens for these operating windows. In steady wind-speed ranges, a strong global forecaster may be sufficient; near a control switch, the same average error can hide whether the model identifies which physical response law is active. Wind-power ramp and transition studies make this operational point directly: sharp changes in output, wake-affected inflow, and turbine control action are exactly where reserve and shortage decisions become exposed [Gallego-Castillo et al., 2015, Khodayar and Wang, 2019, Zhou et al., 2024, Haq et al., 2025]. SCADA-based turbine studies also show why those boundaries are observable only when the relevant sensor fields, such as pitch and power, are present

and reliable [Tautz-Weinert and Watson, 2017]. A parallel problem appears in the lower atmosphere: stable-to-convective transitions reorganize transport and surface forcing over short horizons that matter for subsequent prediction [Hersbach et al., 2020]. This paper treats those transition windows as the accountability target for the router and evaluates the consequence in reserve, violation, and shortage terms.

Spatio-temporal graph models provide the accuracy reference for correlated time series, but they do not by themselves answer the operating-boundary question. Graph propagation and temporal encoding capture asymmetric dependence and dynamic correlation [Li et al., 2018, Yu et al., 2018, Wu et al., 2019, 2020, Guo et al., 2019, Bai et al., 2020], and wind-power variants use spatial structure to represent wake interaction, transport, and turbine coupling [Khodayar and Wang, 2019, Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. These models supply the low-error reference map that any routed model must face. What they do not decide is which local map should own a physical transition when the downstream decision is reserve allocation rather than leaderboard rank.

Mixture-of-experts (MoE) routing gives a natural handle on heterogeneous response laws, but the router must be tied to physics before it can support an engineering interpretation. Classical and modern MoE models allocate inputs to local predictors [Jacobs et al., 1991, Jordan and Jacobs, 1994], while sparse-routing systems expose familiar training failures such as router collapse and expert starvation [Shazeer et al., 2017, Fedus et al., 2022]. Recent sequence-forecasting work imports the same specialization idea into large time-series models [Shi et al., 2025]. Physics-guided learning, in turn, injects prior knowledge through synthetic data, feature construction, architecture design, or physical losses [Karpatne et al., 2017, Read et al., 2019, Park and Park, 2019, Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. The missing step is to make the gate itself accountable for the operating boundary, not only for validation loss.

The gap is at the gate. Regime-transition error is rarely evaluated as a first-class target, although average metrics can hide failure near operating boundaries [Khodayar and Wang, 2019, Wu et al., 2020, Daenens et al., 2025]. Physics-guided models usually regularize latent states or outputs, while the routing decision remains governed by prediction loss [Raissi et al., 2019, Karniadakis et al., 2021, Zehtabiyani-Rezaie et al., 2023]. In control-confounded SCADA streams, that is enough to create a low-error gate with little mechanical meaning. This failure matters because the gate is the part of an MoE that assigns responsibility. If the gate cannot identify the operating boundary, expert specialization becomes a statistical convenience rather than an engineering explanation.

This study puts physical guidance into the routing decision and audits the operating value of doing so. The model first encodes local spatio-temporal context with a directed-diffusion GRU. It then routes each node through a MoE gate that is checked against operating-regime anchors, graph smoothness, wake exposure, and MPPT-to-pitch ambiguity. WTB and ERA5 are paired because they expose different levels of regime observability: WTB hides the aerodynamic boundary inside control action, while ERA5 makes the thermodynamic marker visible through sensible heat flux. The contribution has three parts. First, the paper turns the MoE gate into an operating-boundary accountability object and tests it with replay, intervention, placebo, threshold-validity, and within-WTB spatial/future stress evidence. Second, it measures the cost of that accountability against graph, transformer, persistence, power-curve, gradient-boosted lag, and LTSF-style baselines, rather than presenting the routed model as the lowest-error forecaster. Third, it links the recovered boundary to a frozen validation-to-test reserve diagnostic in which cost, violation, reserve energy, and shortage energy are reported together. The external Kelmarsh/Penmanshiel experiments are included as failure-boundary diagnostics: they show that the WTB operating rule must be re-estimated and tested before it is used outside its source operating envelope.

2 Related Work

2.1 From average wind-power accuracy to transition-window risk

Wind-power forecasting is operationally useful when it supports decisions under variability, not only when it reduces a fleet-wide error score. Reviews and forecasting studies emphasize that wind power is driven by non-stationary weather, wake interaction, turbine coupling, ramp events, and changing control states [Pinson, 2013, Gallego-Castillo et al., 2015, Khodayar and Wang, 2019, Zhou et al., 2024, Haq et al., 2025]. Spatio-temporal graph models address part of this difficulty by encoding dependence across turbines or locations [Li et al., 2018, Yu et al., 2018, Wu et al., 2019, 2020, Guo et al., 2019, Bai et al., 2020], and wind-specific graph models use spatial structure to represent wake interaction and turbine coupling [Park and Park, 2019, Yu et al., 2020, Daenens et al., 2025]. These models define the accuracy baselines for this paper. The open issue is narrower: a low-error graph encoder may still fail to identify the operating transition where reserve and ramp-risk decisions become most sensitive.

2.2 Operating regimes, SCADA observability, and physical accountability

Regime-aware modeling provides a way to separate local response laws, but the separation must be physically accountable. Early adaptive and hierarchical MoE models partitioned inputs among local experts [Jacobs et al., 1991, Jordan and Jacobs, 1994], and modern sparse MoE systems made routing scalable while exposing collapse, starvation, and unstable load allocation as practical failure modes [Shazeer et al., 2017, Fedus et al., 2022]. Sequence-forecasting MoE models import this specialization idea into time-series settings [Shi et al., 2025]. Recent state-aware time-series routing work also points toward exogenous-condition refinement and shared/private expert specialization: ECTO uses exogenous-conditioned regime refinement for ultra-short-term wind-power forecasting [Cao and Wang, 2026], and ARROW uses adaptive rollout plus shared-private MoE routing for multi-scale global weather forecasting [Tian et al., 2026]. These papers motivate stronger alternatives for state-conditioned routing, but they are not baselines in this study. Physics-guided learning offers the complementary principle: theory can enter through features, pretraining, architecture, losses, or post-hoc constraints [Karpatne et al., 2017, Read et al., 2019, Raissi et al., 2019, Karniadakis et al., 2021, Parsa, 2025]. For wind turbines, SCADA observability matters because operating-state labels depend on sensor coverage, pitch channels, availability masks, and turbine-specific power curves [Tautz-Weinert and Watson, 2017, Zhou et al., 2024]. That condition-monitoring and operating-state literature is complementary to this paper: it establishes why state labels can be sensor- and turbine-specific, while our question is how a forecast MoE should use such labels to constrain responsibility near a control boundary. Renewable and atmospheric applications already use physical graphs, lidar information, frequency-domain constraints, and wind-farm power priors to improve physical consistency or interpretability [Park and Park, 2019, Zhang and Zhao, 2021, Li et al., 2025, Zehtabiyani-Rezaie et al., 2023]. This paper moves the prior to the routing layer itself: the gate is evaluated as a responsibility assignment around a mechanism change.

2.3 Forecast-driven reserve allocation and risk diagnostics

Forecast value in power systems is usually realized through reserve, commitment, balancing, and trading decisions. Reserve studies show that variable generation changes operating-reserve requirements and that reserve demand should reflect uncertainty rather than a fixed margin [Doherty and O'Malley, 2005, Ela et al., 2011]. Probabilistic wind forecasting and quantile forecasting provide the statistical bridge from point forecasts to decision risk [Bremnes, 2004, Nielsen et al., 2006, Zhang et al., 2014], while trading and market studies connect wind forecast uncertainty to operating costs and imbalance exposure [Pinson et al., 2007, Wang et al., 2011, Zhou et al., 2013]. Decision-driven forecast evaluation can also use calibrated predictive distributions, scenarios, quantile-regression baselines, and proper scores such as CRPS. This line of work motivates the paper's reserve audit, but our scope is more limited than full market clearing, unit commitment, or a trained probabilistic reserve baseline. We use reserve cost, violation rate, reserve

energy, and shortage energy as a diagnostic lens to test whether an auditable MPPT-to-pitch gate changes the decision tradeoff in the transition window.

2.4 Gap summary

The gap is not a missing high-capacity forecaster. It is the lack of an auditable link between a control-transition boundary, the gate that assigns local responsibility, and the reserve-risk diagnostic that consumes that assignment. Existing graph and physics-guided models provide strong prediction references, and probabilistic wind-power work explains how uncertainty should affect reserve and market decisions. The missing interface is a frozen, boundary-aware diagnostic that asks whether a physically recovered gate changes transition-window cost, violation, reserve energy, and shortage energy without claiming broad site transfer or full dispatch integration.

3 Methodology

3.1 Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The time index is part of the information boundary. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the anchor time t ; the supervised targets begin at $t + 1$. In WTB, active power appears both as a historical SCADA channel and as the future target. The historical/anchor value $\text{Patv}_{i,t}$ may enter the gate because it is observed at the forecast issue time, but $\text{Patv}_{i,t+\tau}$ for $\tau \geq 1$ is used only as the held-out prediction target and its validity mask. The gate is therefore allowed to use the current turbine operating status, not future power. This convention avoids target leakage in the multi-step horizon, but it also narrows the interpretation: without a separate no-Patv or lagged-Patv ablation, the recovered gate should not be read as proving that the router reconstructs the MPPT-to-pitch boundary independently of active-power telemetry.

Table 1 lists the notation used in the routing and loss definitions. Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This is essential because nearby turbines or grid cells can move into different local regimes within the same sequence.

3.2 Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularization. This is the engineering layer of the method: before defining a network, the paper defines what counts as a control regime and when that regime is observable from the data.

3.2.1 WTB operating regimes

In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

Table 1. Core notation used in the routing and loss definitions.

Symbol	Meaning	Dimension
$\mathbf{x}_{i,t}$	Node feature vector at node i , time t	$\mathbb{R}^F$
$\mathbf{X}_{t-H+1:t}$	Input history window	$\mathbb{R}^{H \times N \times F}$
$y_{i,t}$	Scalar target	$\mathbb{R}$
$\bar{p}_{i,t}$	Mean blade-pitch angle (Pab_mean) in WTB	scalar
$\text{Patv}_{i,t}$	Active power observed at anchor time; future values are prediction targets	scalar
$\mathbf{h}_{i,t}$	Encoded node context	$\mathbb{R}^d$
$\mathbf{a}_{i,t}$	Physics anchor vector used by the gate	$\mathbb{R}^q$
$\mathbf{z}_{i,t}$	Gate logits	$\mathbb{R}^E$
$\mathbf{g}_{i,t}$	Gate probabilities after softmax	$\mathbb{R}^E$
$f_e(\cdot)$	Expert- e prediction head	$\mathbb{R}^d \rightarrow \mathbb{R}^P$
$R_{i,t}$	Primary regime anchor label	discrete
$W_{i,t}$	Auxiliary wake flag in WTB	$\{0, 1\}$
$\mathcal{A}_t(i, j)$	Edge weight from node j to node i at time t	scalar
E	Number of experts	scalar
K	Top- K used in the balancing loss	scalar

$$\bar{p}_{i,t} = \frac{1}{3} \left(p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)} \right).$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. This is separate from Patv, the active-power channel and prediction target, and from Wspd, the nacelle wind-speed channel. The regime labels below use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & \text{if } \text{Wspd}_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & \text{if } u_{\text{idle}} \leq \text{Wspd}_{i,t} \leq u_{\text{rated}} \wedge \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & \text{if } \text{Wspd}_{i,t} > u_{\text{rated}} \wedge \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

These thresholds are operating anchors, not universal turbine constants. The reported WTB implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$, with the exact table repeated in the Appendix. They are treated as site- and sensor-specific choices that must be re-estimated before a new-farm interpretation. The threshold audit and local-boundary recalibration evidence test whether the saved gates survive nearby rated-wind and pitch-threshold changes, but they do not exhaust all cut-in, rated-wind, pitch-angle, wake-score, or turbine-control alternatives.

3.2.2 ERA5 thermodynamic regimes

In ERA5, the anchor is built around the stable-to-convective transition. Surface sensible heat flux changes sign across that transition, and large flux gradients mark disturbed periods. Let $\Delta \text{sshf}_{i,t} = \text{sshf}_{i,t} -$

$\text{sshf}_{i,t-1}$. We use a small-margin threshold $\varepsilon_{\text{sshf}}$ for $|\text{sshf}|$. We also use a high-quantile threshold $q_{0.95}^\Delta$ for $|\Delta \text{sshf}|$. The regime label is then defined by

$$R_{i,t}^{\text{era5}} = \begin{cases} 0, & \text{if } \text{sshf}_{i,t} > \varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{convective}), \\ 1, & \text{if } \text{sshf}_{i,t} < -\varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{stable}), \\ 2, & \text{otherwise} \quad (\text{transition/anomalous}). \end{cases}$$

If sensible heat flux is unavailable, the same construction can fall back to an analogous rule based on the temporal gradient of 2 m temperature. The exact thresholds used in the reported experiments are listed in the Appendix.

3.3 Shared architecture

3.3.1 Physical graph construction

The graph is part of the physical problem specification. In WTB, it exposes likely wake pathways. In ER-A5, it preserves geographically local dependence because the thermodynamic regime marker is already visible in the state. Exact graph constants are listed in the Appendix tables.

For WTB, each turbine has a fixed coordinate $\mathbf{p}_i = (x_i, y_i)$. A candidate set is first built by retaining the k_{nn} nearest turbines within a maximum radius d_{max} . Let $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and let $\theta_{i,t}$ denote the local meteorological wind-from direction. The downstream unit vector is

$$\mathbf{u}_{i,t} = \begin{bmatrix} \sin(\theta_{i,t} + \pi) \\ \cos(\theta_{i,t} + \pi) \end{bmatrix}.$$

which gives the streamwise and cross-stream distances

$$d_{ij,t}^{\parallel} = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}, \quad d_{ij,t}^{\perp} = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|.$$

An edge is activated only when turbine j falls inside a downstream cone,

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1} \left[d_{ij,t}^{\parallel} > 0 \wedge \arctan \left(\frac{d_{ij,t}^{\perp}}{\max(d_{ij,t}^{\parallel}, 10^{-6})} \right) \leq \phi \right],$$

and the resulting wake weight is

$$\tilde{\mathcal{A}}_t(i, j) = \exp \left(-\frac{d_{ij,t}^{\parallel}}{\alpha} \right) \exp \left(-\frac{|d_{ij,t}^{\perp}|}{\beta} \right) \mathbb{I}_{ij,t}^{\text{cone}},$$

If wind direction is missing, the implementation falls back to a static proximity weight

$$\mathcal{A}^{\text{static}}(i, j) = \exp \left(-\frac{\|\Delta \mathbf{p}_{ij}\|_2}{d_{\text{max}}} \right).$$

Only the strongest M inbound weights are retained for each target and time step. The same weights define a wake score,

$$s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j),$$

which is later used as an auxiliary routing indicator. Wake interference is handled separately from the MPPT-to-pitch label because it is spatial. Let $q_{0.75}^{\text{wake}}$ denote the upper-quartile threshold of the wake score over valid operating states in the training split. Then

$$W_{i,t} = \begin{cases} 1, & \text{if } s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & \text{if } s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

ERA5 uses a simpler graph because the main uncertainty is local transport over the retained patch. Each grid point is treated as a node with geographic coordinate (φ_i, λ_i) , and the great-circle distance is computed by the Haversine formula

$$d_{ij} = 2R \arctan\left(\frac{\sqrt{a_{ij}}}{\sqrt{1 - a_{ij}}}\right),$$

where $R = 6371$ km and

$$a_{ij} = \sin^2\left(\frac{\varphi_j - \varphi_i}{2}\right) + \cos(\varphi_i) \cos(\varphi_j) \sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right).$$

We then retain a symmetric k_{nn} -nearest-neighbor graph and define

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph. The design is intentionally simpler than WTB because the ERA5 regime marker is already visible in the observed state.

3.3.2 Directed-diffusion GRU encoder and node-level gate

The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)} \mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)} \text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)} \text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)} \mathbf{X}_t^{(\ell)}\right),$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is the observed value at the issue time, equivalent to a status input in a rolling SCADA forecast. It is standardized using training-split statistics and is never filled from the prediction horizon. This makes the router operationally admissible only when current active-power telemetry is available before the forecast is issued. If an operator’s dispatch cutoff does not make current active power available, the same protocol must drop or lag this anchor and rerun the leakage, intervention, and reserve checks before interpreting the gate physically.

3.3.3 Routing stack and evidence boundary

This makes the routing stack an anchor-constrained diagnostic, not an unsupervised operating-state discovery method. In WTB, Wspd and Pab_mean appear both in the gate anchor and in the pseudo-label rule used for alignment, so high gate-regime agreement should be read as evidence that the implemented router obeys a declared operating boundary. It does not prove that the boundary would be recovered from unrelated SCADA channels. The same caution applies to Patv : the current value is an issue-time status input, but without a retrained no- Patv or lagged- Patv ablation the paper cannot separate how much of the gate comes from active-power telemetry versus wind-speed and pitch information. The evidence therefore supports auditable responsibility under the declared anchor set, not anchor-free regime discovery.

3.4 Physics-aligned routing regularizers

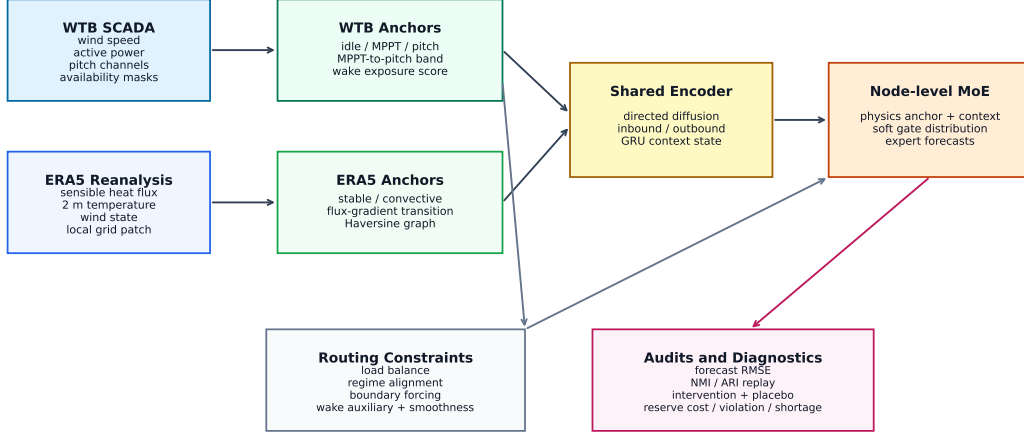
Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [Shazeer et al., 2017, Fedus et al., 2022]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}},$$

with inactive terms set to zero in settings where they are not used. Exact weights are listed in the Appendix.

The training loop is straightforward despite the loss stack. Each mini-batch starts from the history window and the graph sequence, builds the node-level graph context, and encodes the sequence with the directed-diffusion GRU. The gate then combines the encoded context with the physics anchor, produces node-level routing probabilities, and aggregates the expert forecasts. The prediction loss is always evaluated first. Active routing losses then add expert-usage stabilization, regime-anchor alignment, the WTB MPPT/pitch forcing term, wake auxiliary supervision, and graph smoothness only in the settings where those terms are declared. Checkpoint selection remains based on validation RMSE, so the routing penalties constrain responsibility without becoming the validation-selection metric.

Physics-aligned regime-aware MoE for operating-boundary accountability



Interpretation: WTB tests a control-confounded turbine boundary; ERA5 is an observability contrast where the regime marker is visible.
The routed model is evaluated as an accountability object, then linked to a frozen validation-to-test reserve diagnostic.

Figure 1: Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

3.4.1 Prediction objective

The prediction loss remains the anchor of the optimization, because a routing correction that stops serving the forecast defeats the purpose of the model. We retain the masked mean-squared-error loss

$$\mathcal{L}_{\text{pred}} = \frac{\sum_{i,t,\tau} m_{i,t+\tau}^y (\hat{y}_{i,t+\tau} - y_{i,t+\tau})^2}{\sum_{i,t,\tau} m_{i,t+\tau}^y},$$

where $m_{i,t+\tau}^y$ is the target-validity mask. All routing regularizers remain subordinate to this prediction objective.

3.4.2 Load balancing

A slightly favored expert can dominate early training and starve the others before regime-specific structure has time to emerge. A balancing term counteracts that early collapse. It follows the same practical role as Switch-style load and importance penalties [Fedus et al., 2022], but it is written in the notation of node-time routing rather than sequence-token routing. Let $\mathcal{B}$ denote the routed node-anchor samples in a mini-batch after target and anchor masks are applied, and let $B = |\mathcal{B}|$. For the n -th sample in this set, the soft importance of expert e is

$$I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}, \quad P_e = \frac{I_e}{\sum_{r=1}^E I_r}.$$

The empirical top- K load is

$$f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)].$$

The balancing loss couples hard usage and soft probability mass:

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

This statistic is computed per mini-batch over valid node-time samples and averaged by the optimizer over training steps. When the hard Top- K load f_e and the soft share P_e are both close to uniform, $E \sum_e f_e P_e$ is close to one and the penalty is small. The term is therefore a training stabilizer, not a claim about optimal expert allocation or physical ownership. Regime ownership is supplied only by the anchored alignment terms below.

3.4.3 Regime-anchor alignment

Balanced expert usage can still produce a physically meaningless partition. The next term aligns the gate with the coarsest regime structure that can be identified with high confidence. Let $\mathbf{z}_{i,t}^{(1:C)}$ denote the first C gate logits, and let $R_{i,t}$ and $M_{i,t}$ be the primary regime label and its validity mask. The mapping from primary regime labels to the first C logits is fixed by construction: for WTB, idle, MPPT, and pitch-control correspond to logits 1–3; for ERA5, stable, transition, and convective correspond to logits 1–3. This supervised index convention breaks the usual MoE label-permutation symmetry only for the anchored logits. Any extra expert logits remain unassigned unless they are used by the wake auxiliary term. The alignment loss is

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\},$$

with inverse-frequency class weights estimated on the training split. Alignment stabilizes the coarse physical partition only where the labels are reliable. The fixed index convention is an implementation constraint, not a post-hoc naming step: it deliberately breaks permutation symmetry for the anchored logits during training so that responsibility can be audited across seeds. The price is scope. The method cannot claim to discover every alternative data-driven partition that an unconstrained MoE might find; cross-seed statements such as “MPPT-aligned” or “pitch-aligned” refer only to this declared mapping and are not inferred after training by relabeling experts.

3.4.4 Boundary-focused forcing (WTB only)

The MPPT-to-pitch boundary in WTB remains ambiguous even after coarse regime alignment because control action partly masks the mechanical transition. A focused forcing term acts only on the most informative MPPT and pitch-control samples. Let

$$Y_{i,t}^{\text{force}} = \begin{cases} 0, & R_{i,t}^{\text{wtb}} = 1, \\ 1, & R_{i,t}^{\text{wtb}} = 2. \end{cases}$$

Using the MPPT-aligned and pitch-aligned expert logits fixed above, we define

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE}\left(\begin{bmatrix} z_{i,t}^{(\text{mppt})} \\ z_{i,t}^{(\text{pitch})} \end{bmatrix}, Y_{i,t}^{\text{force}}\right),$$

where $\Omega_{\text{force}} = \{(i,t) : R_{i,t}^{\text{wtb}} \in \{1,2\}, M_{i,t} = 1\}$. This term concentrates the intervention on high-confidence MPPT and pitch-control anchors already visible at the issue time. It increases routing identifiability around the boundary after the operating state is expressed in the SCADA channels; it is not designed or evaluated as a lead-time switch predictor.

3.4.5 Wake auxiliary supervision and graph smoothness

After the MPPT-to-pitch boundary is repaired, wake-sensitive samples can still be mixed with non-wake samples inside the same operating regime. A wake auxiliary label identifies that residual ambiguity in WTB, while graph smoothness discourages noisy neighbor-to-neighbor gate jumps in both settings. The wake auxiliary loss is

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples for which the wake flag is defined. The wake flag is a thresholded auxiliary view of the continuous wake score, using the training-split upper quartile reported in the Appendix; the continuous score still enters the anchor vector. This threshold is a pragmatic high-exposure label rather than a complete wake model, and its sensitivity is treated as part of the threshold-limited evidence boundary. The smoothness term is computed on the graph snapshot associated with the anchor time,

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Here $\mathcal{T}_{\mathcal{B}}$ is the set of anchor times represented in the mini-batch, and ϵ prevents division by zero if a graph has no retained edges. The auxiliary term is a WTB-specific repair for wake mixing, while the smoothness term encourages locally coherent routing without prescribing a global partition. This is a bias toward neighbor-consistent responsibility, not a physical law. In disturbed inflow or local-wake conditions, it may smooth over real turbine-level heterogeneity, so wake-specific failures are checked separately rather than treated as solved by graph smoothness.

3.5 Operational reserve diagnostic protocol

The reserve diagnostic is a frozen validation-to-test procedure, not a dispatch simulator or probabilistic forecasting benchmark. Its purpose is to test whether a physically auditable gate changes the reserve tradeoff in the MPPT-to-pitch window after the point-forecast model has already been trained. For each saved WTB run, validation predictions are used only to select reserve levels; test predictions are then evaluated once with those reserve levels fixed. The protocol therefore follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later operating period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ the realized active power at an evaluated horizon cell. We define over-forecast shortfall as

$$s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0),$$

because this is the error direction that leaves a reserve schedule exposed. The quantiles below are empirical quantiles of point-forecast shortfall, not outputs of a trained quantile-regression or distributional forecaster. For a reserve policy b and quantile q , the validation split estimates a reserve level

$$r_{b,q} = Q_q(\{s_{i,t} : (i,t) \in b, (i,t) \in \mathcal{V}\}),$$

where Q_q is an empirical quantile and $\mathcal{V}$ is the validation split. The grid is fixed before test evaluation:

$$q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}.$$

The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{T}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{T}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t.$$

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. All totals are reported in normalized reserve-energy cost units. Thus a value written as 84.58M denotes 84.58 million normalized reserve-energy cost units, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ means that one unit of residual shortage is charged as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware or emergency screening where shortage avoidance dominates local cost savings. The paper evaluates four policies under this protocol: Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin. Only same-model global comparisons are used to attribute gate-bin reserve effects to the learned router. Graph WaveNet/global remains the full-sample low-RMSE system reference, and Graph WaveNet/physical-bin remains the simple physical-stratification baseline that tests how much can be gained without learned gates.

Reserve diagnostic algorithm. Input saved validation/test point forecasts, targets, masks, physical anchors, and gate outputs. Compute over-forecast shortfall on valid cells. Form global, physical-bin, and gate-bin reserve groups on validation outputs only. For each cost ratio and group, select the empirical shortfall quantile that minimizes validation reserve-plus-shortage cost. Freeze the selected reserve levels and evaluate them once on test cells. Aggregate cost, violation rate, reserve energy, and shortage energy over full, boundary, transition, ramp, horizon-bucket, and time-of-day slices. Report horizon-specific empirical-quantile rows only as a post-hoc what-if; no probabilistic model is trained.

4 Experimental Setup

4.1 Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [Zhou et al., 2024]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [Hersbach et al., 2020]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

4.2 In-family comparison protocol

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capac-

Table 2. In-family model comparison used for mechanism validation in the current encoder family.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Tests whether routed capacity alone is enough
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{bal} + L_{align} + L_{force}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Dataset-specific corrected-routing comparator

ity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST are included for WTB, together with deterministic persistence, a fitted physical power-curve baseline, XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split keeps the paper from using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**. The WTB forecasting tables additionally report Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, the full physics-aligned MoE, the WTB boundary-forced router, and engineering baselines built from persistence, power curves, gradient-boosted lag features, and a DLinear-style LTSF readout. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Table 2.

4.3 Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in the Appendix so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST across seeds 201–205, giving 20 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct

from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the physical setting before the forecasting results are introduced.

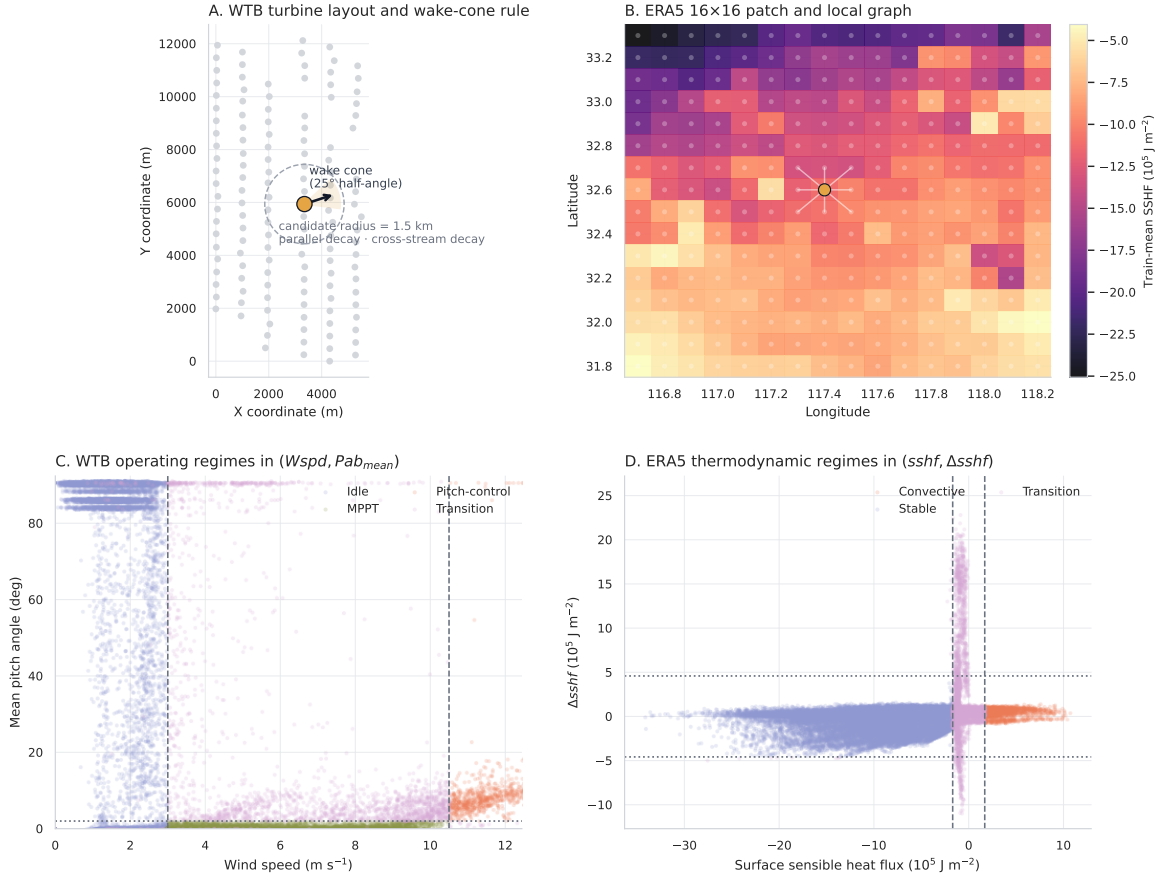


Figure 2: Dataset geometry and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and the retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split.

5 Results

5.1 Overall performance

WTB delivers a hard accuracy result: the strongest graph and engineering forecasters beat the routed family under the synchronized strict-mask refresh. Here and in the baseline tables, “switch-window” refers to the precomputed transition-window mask, not to the rated-wind boundary-band audit below. Graph WaveNet reaches overall RMSE 225.74 $\pm$ 2.60 and switch-window RMSE 228.12 $\pm$ 3.23 across seeds 201–205. PatchTST reaches 228.07 $\pm$ 4.05 and 231.22 $\pm$ 4.15. Graph Transformer reaches 235.38 $\pm$ 5.15 and 237.06 $\pm$ 5.78, while GAT-GRU reaches 236.59 $\pm$ 7.80 and 240.17 $\pm$ 8.50. Among engineering baselines, LightGBM lag features reach overall/switch RMSE 227.12 / 232.24, XGBoost lag features 227.88 / 233.07, a fitted physical power curve 241.02 / 245.37, persistence 243.23 / 246.61,

Table. WTB engineering forecasting baselines evaluated on the strict cache.

Baseline	Overall RMSE	Switch RMSE	Pitch-control RMSE
Persistence	243.23	246.61	262.35
Physical power curve	241.02	245.37	299.30
XGBoost lag-feature	227.88	233.07	277.33
LightGBM lag-feature	227.12	232.24	270.23
DLinear-style LTSF	246.81	251.15	284.06

and the DLinear-style LTSF readout 246.81 / 251.15. The boundary-forced router reaches overall RMSE 236.13 $\pm$ 8.41, switch-window RMSE 239.86 $\pm$ 8.85, and pitch-control RMSE 286.95 $\pm$ 22.07 across seeds 201–205. The completed strict-mask leaderboard places the routed model outside a headline-error objective and motivates the narrower routing-recovery claim.

ERA5 makes the same separation sharper. A last-value persistence baseline gives the lowest headline error, with overall RMSE 0.7127 and switch-window RMSE 0.6814. Among learned ERA5 baselines, Graph WaveNet has the lowest learned-model RMSE at 0.8417 $\pm$ 0.0540, followed by the matched dense encoder at 0.8641 $\pm$ 0.0690, physics-aligned MoE at 0.8735 $\pm$ 0.0479, TCN at 0.8784 $\pm$ 0.0286, PatchTST at 0.8817 $\pm$ 0.0675, Unconstrained MoE at 0.9294 $\pm$ 0.0451, and STGCN at 0.9893 $\pm$ 0.1344. The forecasting tables set the terms of the paper: physics-aligned routing earns its value through interpretable responsibility. That value is only credible if the routing metrics improve under the same evaluation protocol.

The forecast evidence is restricted to the current frozen evaluation split and artifact package. The text reports the completed neural leaderboard, and the table below provides the engineering baseline panel. The corrected-router seed table is reported in the routing-evidence section, where it supports the accountability claim rather than a leaderboard claim.

5.2 Transition-window and regime-slice performance

The difficult slices repeat the same separation between forecasting accuracy and routing evidence. In the completed strict-mask baseline rows, Graph WaveNet reaches switch-window RMSE 228.12 $\pm$ 3.23 and pitch-control RMSE 282.67 $\pm$ 31.41. PatchTST reaches switch-window RMSE 231.22 $\pm$ 4.15 and pitch-control RMSE 277.21 $\pm$ 19.30. Graph Transformer reaches switch-window RMSE 237.06 $\pm$ 5.78 and pitch-control RMSE 280.50 $\pm$ 35.98, while GAT-GRU reaches switch-window RMSE 240.17 $\pm$ 8.50 and pitch-control RMSE 318.28 $\pm$ 49.88. Under the strict anchor-valid mask, the boundary-forced router reaches pitch-control RMSE 286.95 $\pm$ 22.07 and switch-window RMSE 239.86 $\pm$ 8.85. These slice results support a mechanism-evidence update, not a headline-error claim. ERA5 again favors persistence. Among learned ERA5 models, Graph WaveNet gives the lowest overall and switch-window RMSE, while physics-aligned routing improves over Unconstrained MoE in the transition slice without beating the dense model or persistence on headline error. This is calibration around a visible thermodynamic boundary, not broad forecast repair.

The slice evidence is therefore restricted to the frozen strict-mask rows and the boundary-slice audit below.

We therefore add a stricter boundary-slice audit directly from the saved strict-mask predictions and the strict cache physics. The audit isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the rated-wind MPPT-to-pitch boundary and compares them with valid non-boundary MPPT/pitch anchors and with single-regime core slices. The boundary band is the hardest forecasting slice: RMSE is 321.17 $\pm$ 21.31, which is 67.07 higher than the non-boundary valid slice (95% bootstrap interval 52.54 to 84.21), 67.46 higher than the MPPT core, and 50.01 higher than the pitch-control core. At the same time, the boundary band still carries mixed-regime routing structure, with NMI/ARI 0.6562 / 0.7766. This closes a narrower mechanism question: the strict-mask gate is not only globally aligned, but remains measurable at the

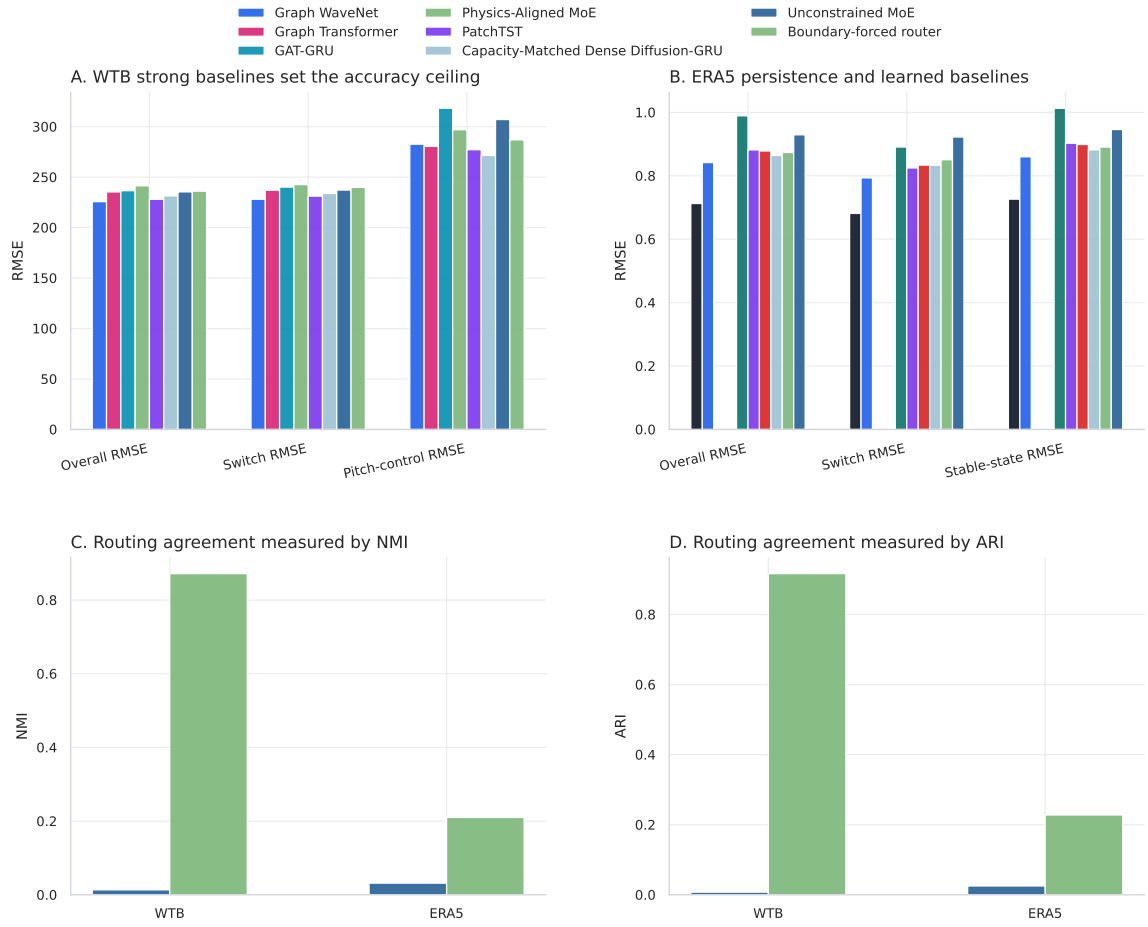


Figure 3: Summary of forecasting and routing outcomes. Panels A and B compress the learned-model error comparison for WTB and ERA5, with ERA5 persistence shown as a deterministic reference. In WTB, the displayed corrected comparator is the boundary-forced routing variant; the deeper full stack is reported separately. Panels C and D summarize routing agreement through NMI and ARI.

Table 1. Strict-mask WTB MPPT-to-pitch boundary-slice audit.

Slice	RMSE	NMI	ARI	Anchor cells
boundary_band	321.1711 +/- 21.3067	0.6562	0.7766	7879
nonboundary_valid	254.0981 +/- 8.6588	0.6326	0.6056	283797
mppt_core	253.7099 +/- 8.7181	NA	NA	277719
pitch_core	271.1653 +/- 16.9980	NA	NA	6175

Table 2. Strict-anchor-mask WTB five-seed metrics.

Seed	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
201	245.0565	246.9875	308.0279	0.8685	0.9242
202	244.3905	248.7028	281.8781	0.9377	0.9692
203	228.0898	229.0400	269.9951	0.8221	0.8658
204	227.7822	232.0819	263.0312	0.8591	0.9060
205	235.3335	242.4639	311.8080	0.8708	0.9177
Mean +/- SD	236.1305 +/- 8.4096	239.8552 +/- 8.8508	286.9480 +/- 22.0652	0.8716 +/- 0.0418	0.9166 +/- 0.0371

operating boundary where prediction is hardest.

5.3 Routing evidence

Routing recovery is the positive result. Across five ERA5 seeds, Unconstrained MoE keeps high expert-usage entropy but near-zero routing agreement, with NMI/ARI of 0.0317 +/- 0.0219 / 0.0253 +/- 0.0626. Physics-aligned MoE raises those scores to 0.2100 +/- 0.0972 / 0.2279 +/- 0.1296. This is a modest but useful correction because the ERA5 marker is already visible in the input state. WTB shows the stronger effect after the strict anchor-valid rerun: the boundary-forced router reaches gate-regime NMI/ARI of 0.8716 +/- 0.0418 / 0.9166 +/- 0.0371 across seeds 201–205. The previous weak seed-204 behavior disappears under the strict cache, with seed 204 reaching NMI/ARI 0.8591 / 0.9060. Explicit gate correction therefore recovers a physically meaningful partition in the control-confounded setting, while the result remains bounded to the tested WTB/ERA5 settings.

Anchor-only routing is not a straw baseline. A deterministic rule built from the physical anchor reaches slightly higher strict-test semantic agreement than the trainable boundary-forced router (NMI/ARI 0.8895/0.9219 versus 0.8716/0.9166), and it also has lower headline RMSE. The proposed model is therefore not claimed to be a better semantic rule in isolation. Its advantage is narrower: it keeps trainable MoE responsibility assignment coupled to the forecast and reserve evaluation, lowers strict pitch-control RMSE relative to anchor-only, and is much less brittle when boundary-anchor channels are corrupted. Under 0.5-standard-deviation boundary-anchor noise, anchor-only overall RMSE degradation is 105.56, whereas the trainable router degrades by 5.78. This is the answer to the rule-based-router concern: if a site only needs a fixed labeler, anchor-only is strong; if the gate must remain part of a forecast/reserve model under sensor noise and held-out turbine stress, trainable responsibility matters.

The current WTB mechanism table above is the evidence source for the corrected router. ERA5 routing values are used only as the observability contrast already summarized in the text and artifact package.

Figure 4 makes the WTB repair visible in the operating plane. After correction, the dominant expert regions align with the MPPT and pitch-control structure, and the confusion matrices contract around the intended operating partition. The corrected gate becomes readable at the boundary that matters operationally, with the RMSE cost already quantified above.

Figure 5 traces the same contrast through time. In the WTB switch window, the boundary-forced router carries responsibility through the MPPT-to-pitch transition more coherently than the unconstrained gate.

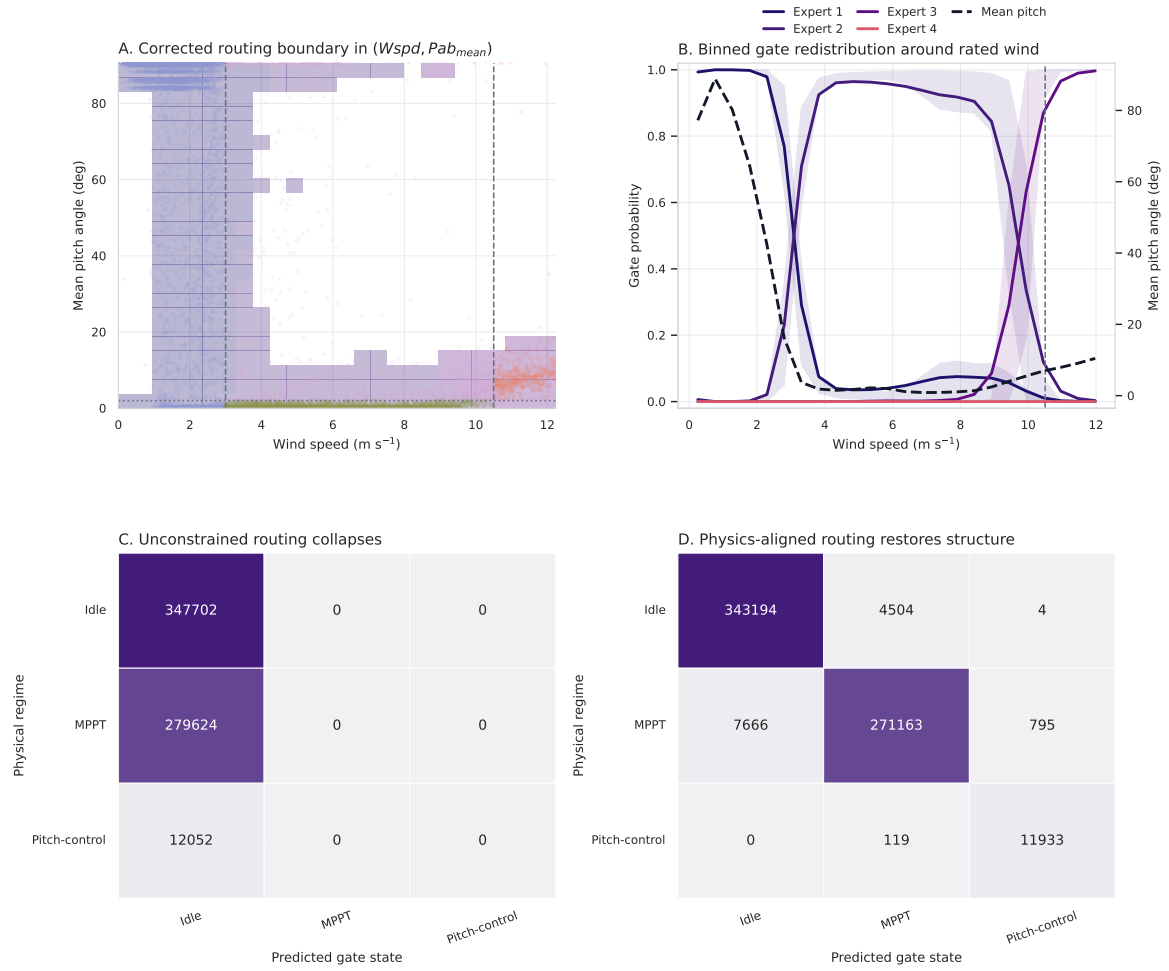


Figure 4: WTB routing diagnostics. Panel A maps the dominant expert in the $(Wspd, Pab_{mean})$ plane for the boundary-forced routing variant, with the sample cloud colored by physical operating regime. Panel B shows the binned gate redistribution against wind speed together with the mean pitch curve. Panels C and D compare the unconstrained and corrected confusion matrices.

Table. Anchor-only and rule-based router comparison on strict-mask WTB.

Model	Overall	Pitch	NMI/ARI	Noise degrade	Reading
MoE + L_bal + L_align + L_force	236.13	286.95	0.8716/0.9166	5.78	trainable responsibility and reserve coupling
Anchor-only router	233.46	317.81	0.8895/0.9219	105.56	strong semantic rule baseline
Context-supervised router	242.75	287.75	0.7521/0.8020	NA	accurate but less boundary-accountable control
Physics-Aligned MoE	241.42	297.02	0.8312/0.8820	NA	routed forecasting compromise

In ERA5, expert weights evolve with sensible heat flux and regime shading, as expected when the marker is already expressed in the state. The case studies tie the routing metrics to the physical mechanisms behind them.

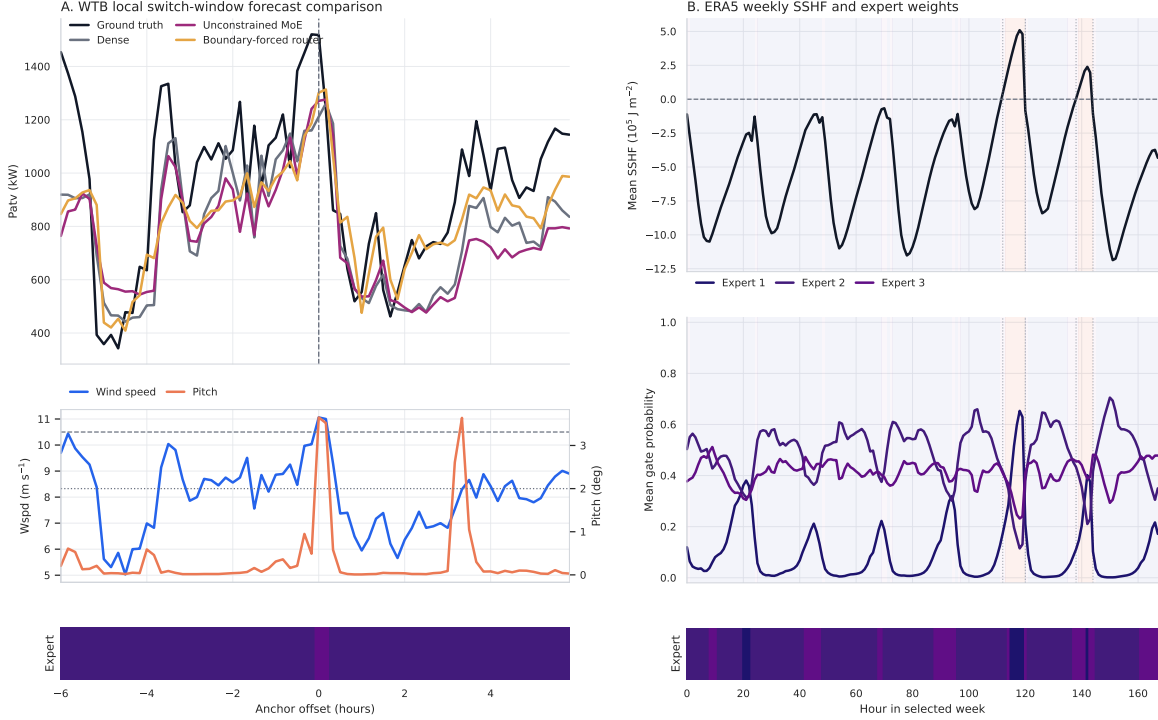


Figure 5: Dual case studies for correction and emergence. Panel A shows a local WTB switch window in which the boundary-forced routing variant remains stable while the dense and unconstrained models deviate more strongly near the MPPT-to-pitch transition; wind speed, pitch angle, and the dominant expert strip are plotted underneath. Panel B shows the ERA5 positive-control view, where sensible heat flux, regime shading, and expert weights evolve coherently over a selected week.

5.4 Operational value and failure cases

The routing claim needs an energy-system consequence, not only a semantic score. A high gate NMI is useful only if it changes the tradeoff an operator actually sees in a transition window: how much reserve energy is carried, how often the scheduled reserve is violated, and how much shortage energy remains when the turbine crosses from MPPT into pitch control. The reserve diagnostic defined in the Method section applies that test to saved WTB validation and test predictions. It reports total cost, violation rate, reserve energy, and shortage energy across shortage-to-reserve cost ratios for four policies: Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin. The paired uncertainty check below is deliberately conservative and does not support a stable whole-sample reserve advantage over Graph WaveNet/global with five paired seeds. The reserve result should therefore be read through the boundary-window tradeoff, not as a full-sample policy win. Because the reserve levels are empirical shortfall quantiles from point forecasts, the table is also not a comparison to

calibrated probabilistic, CRPS-scored, quantile-regression, scenario, or distributional reserve baselines.

These four policies answer different decisions and should not be ranked as substitute forecasters. Graph WaveNet/global is the low-RMSE system reference: one reserve quantile is calibrated for the whole sample, so it tests what a strong black-box forecaster can do without boundary stratification. Graph WaveNet/physical-bin keeps the same strong forecaster but calibrates reserve by physical operating bins, answering whether a simple physics-stratified reserve rule already captures the transition-window effect. Boundary router/global uses the higher-RMSE routed model with one global reserve quantile, isolating the cost of using the accountable router without gate-specific reserve allocation. Boundary router/gate-bin adds gate-conditioned reserve bins to that same routed model, so its only legitimate attribution question is whether the learned gate improves the same routed predictor’s transition-window reserve tradeoff relative to its own global reserve rule.

At the main shortage-to-reserve cost ratio of 10, the full-sample result still favors Graph WaveNet/global as the system reference: its total cost is 464.07M normalized reserve-energy cost units with RMSE 225.74. The boundary-forced router pays a 10.39 RMSE penalty. Gate-bin reserve does not erase that penalty and does not become the preferred full-sample reserve policy: on the full sample it lowers same-model cost by carrying less reserve, but violation rises from 0.0857 to 0.1214 and shortage rises from 17.64M to 19.02M. The useful result is local and has the form of a reserve envelope. On boundary anchors, the boundary router/gate-bin policy lowers same-model total cost from 88.13M to 84.58M, violation from 0.1038 to 0.0900, and shortage from 3.73M to 3.19M, while reserve energy rises from 50.82M to 52.67M. The operational gain is therefore a 0.0138 absolute violation reduction and 0.54M lower boundary shortage, bought with 1.85M additional reserve energy. Graph WaveNet/physical-bin remains the strongest physics-stratified boundary baseline at 84.31M total cost and 3.40M shortage. Thus the gate-bin result is not an accuracy result; it is a boundary-window reserve allocation that buys lower shortage and fewer violations at a measured reserve-energy price.

Table. Dispatch/reserve main table at shortage-to-reserve cost ratio 10.

Subset	Policy	RMSE penalty	Q	Cost	Violation	Reserve	Shortage	Reading
boundary	Graph WaveNet/physical-bin	+0.00	low=0.900, mid=0.900, high=0.900	84.31M	0.0931	50.35M	3.40M	physics-stratified graph reference
boundary	Boundary router/gate-bin	+10.39	low=0.900, mid=0.900, high=0.900	84.58M	0.0900	52.67M	3.19M	wins on risk, pays reserve
boundary	Boundary router/global	+10.39	all=0.900	88.13M	0.1038	50.82M	3.73M	same-router global reference
boundary	Graph WaveNet/global	+0.00	all=0.900	88.80M	0.1022	50.32M	3.85M	system reference
full	Graph WaveNet/global	+0.00	all=0.900	464.07M	0.0901	288.01M	17.61M	system reference
full	Graph WaveNet/physical-bin	+0.00	low=0.900, mid=0.900, high=0.900	464.43M	0.1057	278.15M	18.63M	physics-stratified graph reference
full	Boundary router/gate-bin	+10.39	low=0.900, mid=0.900, high=0.900	481.36M	0.1214	291.16M	19.02M	cost-only tradeoff
full	Boundary router/global	+10.39	all=0.900	501.32M	0.0857	324.97M	17.64M	same-router global reference

The cost ratio gives the reserve audit its energy-system meaning. A ratio of 2 means that one unit of residual shortage is charged as two units of reserve energy, so the optimizer tolerates more shortage and selects the median shortfall quantile. Ratios 5 and 10 represent a moderate reliability preference, such as transition-window scheduling where shortage or imbalance is expensive enough to justify extra reserve but not so expensive that the safest global rule dominates every local allocation. Ratios 20 and 50 represent scarcity-aware dispatch, high imbalance penalties, or emergency reliability screening where the decision rule must prioritize very low violation and shortage rates over local cost savings. Under this reading, the win/loss boundary is explicit. At ratio 2, the selected reserve quantile is 0.50 and the gate-bin boundary policy does not buy reserve. At ratios 5 and 10, gate-bin is usable relative to the same model’s global reserve because it reduces boundary shortage and violation while carrying more reserve. At ratio 20, the benefit narrows into a cost-only tradeoff: shortage falls slightly but violation is effectively unchanged. At ratio 50, gate-bin falls outside the acceptable region because the same-model global policy is safer and cheaper. The gate is therefore a conditional reserve-allocation signal around the MPPT-to-pitch boundary, not a replacement for unit commitment, market clearing, or system-level stochastic reserve optimization.

The validation-selected quantiles are also checked on held-out coverage. At ratio 10, all four displayed policies select an average target coverage near 0.90, but realized coverage differs by subset and policy. The boundary router/gate-bin policy is closest to the target on boundary anchors, with realized coverage

Table. Energy-system interpretation of shortage-to-reserve cost ratios.

Ratio	System assumption	Economic mapping	Use in paper
2	low scarcity or curtailment-tolerant screening	if reserve costs C_r per MWh, shortage is charged $2 \cdot C_r$	diagnostic only; reserve quantile stays near the median
5	moderate reliability preference for transition-window scheduling	if reserve costs C_r per MWh, shortage is charged $5 \cdot C_r$	usable same-model boundary tradeoff if violation and shortage fall
10	main operating case: shortage materially more expensive than reserve	if reserve costs C_r per MWh, shortage is charged $10 \cdot C_r$	primary reserve diagnostic; not a market-clearing claim
20	scarcity-aware dispatch or high imbalance penalty	if reserve costs C_r per MWh, shortage is charged $20 \cdot C_r$	narrow cost-only region; requires ramp-specific checks
50	emergency reliability or VOLL-like shortage avoidance	if reserve costs C_r per MWh, shortage is charged $50 \cdot C_r$	gate-bin reserve not supported; global policy is safer

Table. Held-out coverage check for validation-selected reserve quantiles at shortage-to-reserve cost ratio 10.

Subset	Policy	Target cov.	Realized cov.	Error	Violation	Shortage	Reading
boundary	Boundary-forced router/gate-bin	0.900	0.910	+0.010	0.0900	3.19M	near calibration target
boundary	Boundary-forced router/global	0.900	0.896	-0.004	0.1038	3.73M	near calibration target
boundary	Graph WaveNet/global	0.900	0.898	-0.002	0.1022	3.85M	near calibration target
boundary	Graph WaveNet/physical-bin	0.900	0.907	+0.007	0.0931	3.40M	near calibration target
full	Boundary-forced router/gate-bin	0.900	0.879	-0.021	0.1214	19.02M	under-covers on held-out test
full	Boundary-forced router/global	0.900	0.914	+0.014	0.0857	17.64M	near calibration target
full	Graph WaveNet/global	0.900	0.910	+0.010	0.0901	17.61M	near calibration target
full	Graph WaveNet/physical-bin	0.900	0.894	-0.006	0.1057	18.63M	near calibration target

0.910 and violation 0.0900, whereas its full-sample realized coverage falls to 0.879 and exposes the same full-sample risk noted above. This table is a reliability diagnostic for the reserve rule; it is not a calibrated distributional forecast, and it does not replace comparisons to quantile-regression or scenario-based reserve methods.

Two post-hoc sensitivity checks make the reserve boundary more explicit. First, the boundary router/gate-bin rule is re-read on cell-level horizon and time-of-day slices using validation-frozen empirical shortfall quantiles. The late horizon bucket carries the largest residual shortage and violation rate, and the 12:00–18:00 anchor-index proxy bucket is also riskier than the night/evening buckets. The time-of-day labels use the WTB 10 min anchor index modulo 144 rather than a farm-local market clock, so they are a diagnostic proxy only. Second, a horizon-specific empirical-quantile what-if recalibrates empirical quantiles by horizon bucket without training a quantile model. It lowers residual shortage in both Graph WaveNet and routed rows, which shows that the pooled-horizon reserve rule is sensitive to lead-time calibration. This supports adding trained quantile-regression, CRPS-scored distributional, or scenario reserve baselines in future work; it does not strengthen the main gate-bin policy claim.

The same result can be read as a system-value envelope. The gate-bin policy is reserve-active only when the shortage penalty is high enough to make boundary shortages matter but not so high that the safer global policy dominates. In this experiment, that usable range is the moderate cost-ratio interval 5–10 because both endpoints show the same operational pattern: fewer boundary violations and less boundary shortage, bought with a finite increase in reserve energy. At ratio 5, the gate-bin policy lowers same-model boundary cost by 1.95M, violation by 0.0170, and shortage by 0.82M, while adding 2.17M reserve energy. At ratio 10, it lowers same-model boundary cost by 3.55M, violation by 0.0138, and shortage by 0.54M, while adding 1.85M reserve energy. At ratio 20 the benefit is no longer a clear reliability gain, and at ratio 50 the gate-bin policy is outside the acceptable region because it raises cost, violation, and shortage relative to the same-model global policy. This is the system claim: the gate marks a transition-window reserve tradeoff region, not a universally preferred reserve controller.

Table. Boundary-window reserve value envelope for the gate-bin policy. Positive reserve means additional reserve energy relative to the same routed model with a global reserve rule.

Ratio	Use window	Cost vs global	Violation vs global	Extra reserve	Shortage vs global	Vs graph bin	System reading
2	not reserve-active	+0.00M	+0.0000	+0.00M	+0.00M	+1.21M	Shortage penalty is too low to activate reserve.
5	usable moderate-cost window	-1.95M	-0.0170	+2.17M	-0.82M	-0.12M	Boundary gate is useful only when fewer violations/shortages justify extra reserve.
10	usable moderate-cost window	-3.55M	-0.0138	+1.85M	-0.54M	+0.26M	Boundary gate is useful only when fewer violations/shortages justify extra reserve.
20	narrow cost-only window	-1.39M	+0.0001	-0.70M	-0.03M	+1.49M	Benefit narrows; use only after ramp-specific checks.
50	not applicable at high penalty	+5.93M	+0.0035	-1.04M	+0.14M	+6.95M	Same-model global policy is safer at this penalty.

We also export operator-facing slices that are closer to reserve desk language than the aggregate boundary/non-boundary split. The four slices are MPPT-to-pitch transitions, pitch-to-MPPT transitions, validation-calibrated high absolute ramp, and validation-calibrated low absolute ramp. For the boundary-

Table. Post-hoc horizon and time-of-day reserve sensitivity for the boundary router/gate-bin policy at shortage-to-reserve cost ratio 10.

Slice	Bucket	Policy	Cost	Violation	Reserve	Shortage	Cells
horizon	h01_h06	Boundary router/gate-bin	21.62M	0.0411	18.53M	0.31M	353950
horizon	h07_h12	Boundary router/gate-bin	24.20M	0.0785	18.47M	0.57M	352927
horizon	h13_h24	Boundary router/gate-bin	65.69M	0.1422	36.28M	2.94M	693326
time_of_day	tod_00_06	Boundary router/gate-bin	44.88M	0.0967	29.21M	1.57M	563849
time_of_day	tod_06_12	Boundary router/gate-bin	26.08M	0.1085	17.22M	0.89M	328791
time_of_day	tod_12_18	Boundary router/gate-bin	16.19M	0.1807	8.52M	0.77M	160400
time_of_day	tod_18_24	Boundary router/gate-bin	24.37M	0.0623	18.32M	0.60M	347163

Table. Horizon-specific empirical-quantile what-if. This recalibrates empirical shortfall quantiles by horizon bucket without training a probabilistic forecaster.

Subset	Policy	Pooled cost	Horizon-Q cost	Delta violation	Delta reserve	Delta shortage	Reading
boundary	Boundary router/gate-bin	111.51M	107.03M	-0.0045	+0.79M	-0.53M	risk improves, post-hoc only
boundary	Boundary router/global	112.12M	107.74M	-0.0055	+1.24M	-0.56M	risk improves, post-hoc only
boundary	Graph WaveNet/global	111.27M	106.76M	-0.0054	+0.91M	-0.54M	risk improves, post-hoc only
boundary	Graph WaveNet/physical-bin	109.83M	104.27M	-0.0032	-0.36M	-0.52M	risk improves, post-hoc only
full	Boundary router/gate-bin	516.49M	494.01M	-0.0002	-13.17M	-0.93M	risk improves, post-hoc only
full	Boundary router/global	531.00M	514.86M	-0.0042	-7.39M	-0.88M	risk improves, post-hoc only
full	Graph WaveNet/global	492.78M	469.99M	-0.0056	-15.97M	-0.68M	risk improves, post-hoc only
full	Graph WaveNet/physical-bin	487.75M	460.81M	-0.0040	-20.09M	-0.68M	risk improves, post-hoc only

forced router, gate-bin reserve reduces violation and shortage energy in the MPPT-to-pitch slice relative to the same model’s global policy, but it pays an added reserve-cost price: 17.93M total cost, 0.1027 violation rate, and 0.69M shortage energy, compared with 16.33M, 0.1124, and 0.77M under global reserve. The low-ramp slice shows the cleaner cost/risk improvement, with gate-bin total cost 10.73M versus 11.24M and shortage energy 0.40M versus 0.44M. The pitch-to-MPPT and high-ramp slices are not gate-bin wins. In the high-ramp slice, gate-bin reserve reduces total cost slightly but raises violation from 0.0766 to 0.1253 and shortage energy from 1.06M to 2.51M because it carries much less reserve energy. This is why the reserve claim remains a boundary-window decision diagnostic rather than a global reserve-policy claim.

The operational decision curve puts the RMSE penalty and reserve-risk tradeoff in one view. It overlays boundary-window shortage energy across cost ratios with the boundary router/gate-bin reserve energy and its fixed RMSE penalty relative to Graph WaveNet/global. The higher-RMSE routed model is considered only for a transition-window reserve decision and only inside the envelope where fewer boundary shortages and violations are worth the extra reserve. Outside that envelope, the lower-RMSE Graph WaveNet/global or Graph WaveNet/physical-bin policy remains the more appropriate system reference.

The three operational cases in the next table are included to prevent a common misreading. A correct gate can help allocate attention around the MPPT-to-pitch boundary, but it can also be misleading when the downstream forecast is bad or when a low-reserve gate-bin policy is applied to high-ramp windows.

The paired uncertainty table is intentionally conservative. It compares full-sample reserve outcomes against Graph WaveNet/global using seed-paired differences with bootstrap intervals and a sign-permutation check. These rows do not show a stable whole-sample cost advantage for the routed policies; the intervals are wide with five seeds. The reserve benefit is therefore the boundary-window operational pattern above, while the whole-sample comparison remains a diagnostic check.

Lead-lag analysis gives the same operational boundary. Around physical regime switches, the corrected gate matches the new regime at rate 0.4066 six steps before the switch and 0.3636 three steps before it, then rises to 0.8008 at the switch and remains 0.6313/0.6094 three/six steps after it. This pattern supports state detection at the transition, not a lead-time alarm. The reserve experiment should therefore be

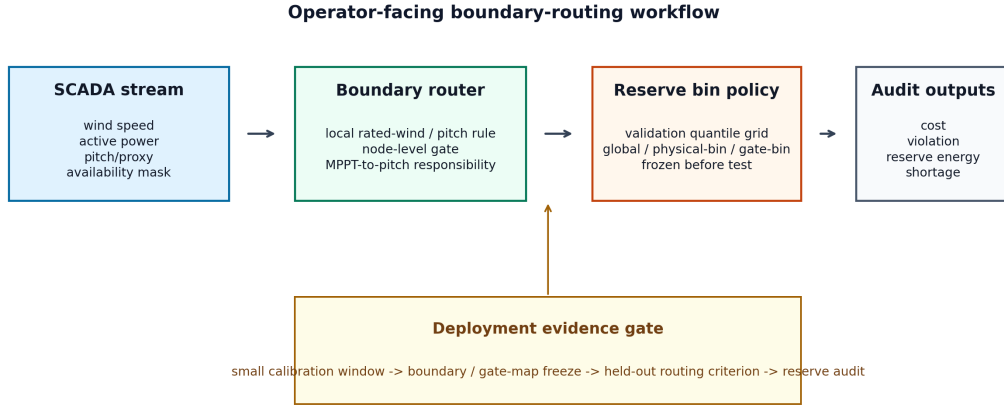


Figure 6: Operator-facing workflow for the reserve audit. SCADA fields define the local operating boundary, the boundary router assigns MPPT-to-pitch responsibility, the reserve-bin policy is calibrated on validation shortfall and frozen before test, and the reported outputs are cost, violation, reserve energy, and shortage energy.

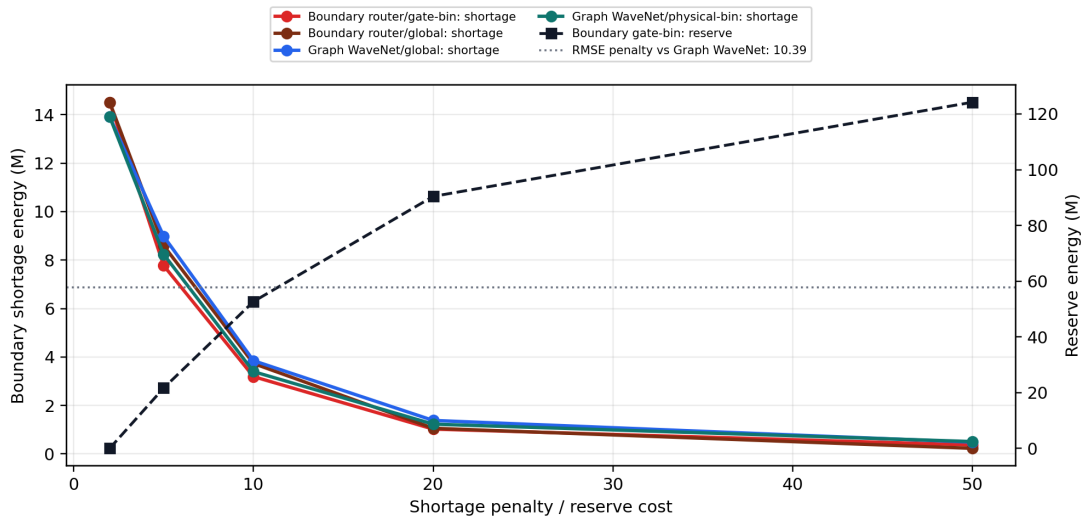


Figure 7: Operational decision curve for the WTB boundary window. The chart places the boundary router's RMSE penalty against Graph WaveNet/global beside reserve energy and shortage-energy changes across shortage-to-reserve cost ratios. The figure explains the restricted decision logic: the routed model is not chosen for average error, but examined for a boundary-window reserve tradeoff that is acceptable only in the moderate cost-ratio envelope.

Table. Operator-facing cases: when the boundary gate helps and when it misleads.

Case	Observed	Operator reading	Boundary reading
MPPT-to-pitch reserve	global cost 16.33M, violation 0.1124, shortage 0.77M; gate-bin cost 17.93M, violation 0.1027, shortage 0.69M	gate-bin lowers transition-window violation and shortage but costs more reserve	boundary-window diagnostic, not global reserve optimization
high-ramp reserve	global violation 0.0766, shortage 1.06M, reserve 57.03M; gate-bin violation 0.1253, shortage 2.51M, reserve 42.02M	gate-bin can look cheaper by carrying too little reserve	failure slice; require ramp-specific checks
gate-correct but forecast-bad	model MoE + L _{bul} + L _{align} + L _{force} , seed 203, gate_correct 1, RMSE 1258.50	right responsibility assignment does not guarantee an accurate value map	mechanism accountability, not dispatch-ready point forecasting

Table. Boundary-window operational reserve slices at shortage-to-reserve cost ratio 10.

Slice	Policy	Cost	Violation	Reserve	Shortage	Valid cells
mppt_to_pitch	Boundary-forced router/global	16.33M	0.1124	8.62M	0.77M	1968
mppt_to_pitch	Boundary-forced router/physical-bin	16.36M	0.0936	10.72M	0.56M	1968
mppt_to_pitch	Graph WaveNet/physical-bin	17.43M	0.0981	10.83M	0.66M	1968
mppt_to_pitch	Graph WaveNet/global	17.55M	0.1109	9.33M	0.82M	1968
mppt_to_pitch	Boundary-forced router/gate-bin	17.93M	0.1027	11.02M	0.69M	1968
pitch_to_mppt	Boundary-forced router/global	20.27M	0.0604	17.38M	0.29M	1992
pitch_to_mppt	Boundary-forced router/physical-bin	20.65M	0.0772	17.16M	0.35M	1992
pitch_to_mppt	Boundary-forced router/gate-bin	20.83M	0.0833	16.72M	0.41M	1992
pitch_to_mppt	Graph WaveNet/global	21.26M	0.0982	15.49M	0.58M	1992
pitch_to_mppt	Graph WaveNet/physical-bin	21.59M	0.1047	15.21M	0.64M	1992
high_ramp	Graph WaveNet/physical-bin	65.83M	0.1224	45.76M	2.01M	6212
high_ramp	Boundary-forced router/gate-bin	67.08M	0.1253	42.02M	2.51M	6212
high_ramp	Graph WaveNet/global	67.15M	0.0965	51.67M	1.55M	6212
high_ramp	Boundary-forced router/global	67.67M	0.0766	57.03M	1.06M	6212
high_ramp	Boundary-forced router/physical-bin	68.02M	0.1065	49.96M	1.81M	6212
low_ramp	Boundary-forced router/gate-bin	10.73M	0.0845	6.70M	0.40M	5594
low_ramp	Graph WaveNet/physical-bin	10.84M	0.0822	6.85M	0.40M	5594
low_ramp	Graph WaveNet/global	10.88M	0.0817	6.95M	0.39M	5594
low_ramp	Boundary-forced router/physical-bin	11.15M	0.0837	6.79M	0.44M	5594
low_ramp	Boundary-forced router/global	11.24M	0.0849	6.84M	0.44M	5594

framed as boundary-aware reserve/ramp-risk conditioning after the boundary is expressed in the state.

The reviewer-statistics pack reports the complementary failure mode. It contains 24 gate-correct high-error cases and 24 boundary gate-correct high-error cases, and the full lists are retained in the supplement. These rows are important because they separate the two parts of the model. In those samples, the router assigns the sample to the physically appropriate operating regime, but the expert forecast is still numerically poor. The usual pattern is a value-map failure near high-error boundary windows, not a collapse of mechanism responsibility. We therefore treat these cases as evidence for the paper’s boundary condition: correct routing is necessary for mechanism interpretation, but it is not sufficient for accurate power prediction.

5.5 Spatial and future holdout stress tests

The spatial holdout tests whether the routing evidence survives when entire turbines are unavailable during training and validation. The east-node protocol holds out 27 of 134 turbines (20.15%). During training and validation, target and regime-valid entries for holdout nodes are masked; during testing, train-node targets are masked and evaluation is restricted to the held-out nodes. The protocol records `mask_policy_pass = true`, zero strict-anchor violations, 15/15 completed runs, and a passed mechanism gate. In this harder within-farm spatial transfer setting, the boundary-forced router reaches overall RMSE 233.87 +/- 2.84, switch-window RMSE 234.87 +/- 2.33, and NMI/ARI 0.8340 +/- 0.0958 / 0.8829 +/- 0.0875. Context-supervised routing is more accurate but less semantically aligned, with NMI/ARI 0.7479 +/- 0.0461 / 0.8084 +/- 0.0717. Anchor-only routing is semantically strong but less stable. The spatial intervention audit keeps the same interpretation as the in-sample strict-mask mechanism test: boundary-anchor removal causes a large NMI/ARI loss and a positive RMSE penalty, while wake-anchor removal remains near-null. This is not an out-of-site wind-farm result; it is a within-farm node-held-out

Table. Boundary-window cost-ratio sensitivity in reserve-system language.

Ratio	Policy	Q	Cost	Violation	Reserve	Shortage	Vs global
2	Boundary router/gate-bin	low=0.500, mid=0.500, high=0.500	29.02M	0.3420	0.00M	14.51M	not a win
2	Boundary router/global	all=0.500	29.02M	0.3420	0.00M	14.51M	reference
2	Graph WaveNet/global	all=0.500	27.82M	0.3237	0.00M	13.91M	reference
2	Graph WaveNet/physical-bin	low=0.500, mid=0.500, high=0.500	27.82M	0.3237	0.00M	13.91M	not a win
5	Boundary router/gate-bin	low=0.800, mid=0.500, high=0.800	60.50M	0.2029	21.62M	7.77M	wins on risk, pays reserve
5	Boundary router/global	all=0.800	62.44M	0.2199	19.45M	8.60M	reference
5	Graph WaveNet/global	all=0.800	62.61M	0.2255	17.69M	8.98M	reference
5	Graph WaveNet/physical-bin	low=0.800, mid=0.800, high=0.800	60.62M	0.2043	19.42M	8.24M	wins on risk, pays reserve
10	Boundary router/gate-bin	low=0.900, mid=0.900, high=0.900	84.58M	0.0900	52.67M	3.19M	wins on risk, pays reserve
10	Boundary router/global	all=0.900	88.13M	0.1038	50.82M	3.73M	reference
10	Graph WaveNet/global	all=0.900	88.80M	0.1022	50.32M	3.85M	reference
10	Graph WaveNet/physical-bin	low=0.900, mid=0.900, high=0.900	84.31M	0.0931	50.35M	3.40M	wins on risk, pays reserve
20	Boundary router/gate-bin	low=0.950, mid=0.950, high=0.950	110.90M	0.0375	90.41M	1.02M	cost-only tradeoff
20	Boundary router/global	all=0.950	112.30M	0.0374	91.11M	1.06M	reference
20	Graph WaveNet/global	all=0.950	113.09M	0.0443	85.44M	1.38M	reference
20	Graph WaveNet/physical-bin	low=0.950, mid=0.950, high=0.950	109.41M	0.0411	84.84M	1.23M	operational win
50	Boundary router/gate-bin	low=0.975, mid=0.975, high=0.975	142.73M	0.0160	124.20M	0.37M	risk loss
50	Boundary router/global	all=0.975	136.80M	0.0125	125.24M	0.23M	reference
50	Graph WaveNet/global	all=0.975	136.90M	0.0236	112.61M	0.49M	reference
50	Graph WaveNet/physical-bin	low=0.975, mid=0.975, high=0.975	135.78M	0.0234	110.26M	0.51M	cost-only tradeoff

Table. Paired uncertainty for full-sample reserve metrics versus Graph WaveNet/global.

Candidate	Metric	Delta	95% CI	P(lower)	Perm p
Graph WaveNet/physical-bin	total_cost	+0.36M	[-1.14M, +2.28M]	0.400	0.810
Graph WaveNet/physical-bin	violation_rate	+0.0156	[+0.0143, +0.0168]	0.000	0.065
Graph WaveNet/physical-bin	reserve_energy	-9.87M	[-10.86M, -8.83M]	1.000	0.065
Graph WaveNet/physical-bin	shortage_energy	+1.02M	[+0.80M, +1.28M]	0.000	0.065
Physics-Aligned MoE/gate-bin	total_cost	+30.67M	[-63.36M, +111.49M]	0.400	0.629
Physics-Aligned MoE/gate-bin	violation_rate	+0.0251	[+0.0098, +0.0391]	0.200	0.127
Physics-Aligned MoE/gate-bin	reserve_energy	+9.07M	[-14.62M, +30.47M]	0.200	0.498
Physics-Aligned MoE/gate-bin	shortage_energy	+2.16M	[-5.94M, +9.57M]	0.400	0.685
Boundary-forced router/global	total_cost	+37.25M	[-31.51M, +107.97M]	0.400	0.380
Boundary-forced router/global	violation_rate	-0.0044	[-0.0198, +0.0122]	0.600	0.631
Boundary-forced router/global	reserve_energy	+36.95M	[+21.02M, +56.35M]	0.000	0.065
Boundary-forced router/global	shortage_energy	+0.03M	[-5.77M, +6.74M]	0.600	1.000
Boundary-forced router/gate-bin	total_cost	+17.28M	[-52.09M, +87.95M]	0.600	0.752
Boundary-forced router/gate-bin	violation_rate	+0.0312	[+0.0151, +0.0529]	0.000	0.065
Boundary-forced router/gate-bin	reserve_energy	+3.15M	[-10.24M, +17.86M]	0.400	0.628
Boundary-forced router/gate-bin	shortage_energy	+1.41M	[-4.59M, +8.42M]	0.600	0.752

stress test that checks whether the learned routing semantics extrapolate to unseen turbines.

The future-period holdout uses a pre-specified later WTB period with seeds 301–305 and evaluates only the holdout split. All 5/5 runs complete, the mechanism gate passes, and the downstream holdout mechanism intervention, placebo, boundary-slice, and reviewer statistics packs are present. The boundary-forced router reaches holdout overall RMSE 235.49 +/- 7.19, switch-window RMSE 239.93 +/- 7.27, and NMI/ARI 0.8352 +/- 0.0669 / 0.8870 +/- 0.0669. This pre-specified future holdout is the main temporal stress evidence. The later post-hoc time-forward slice in the next robustness block is used only as failure analysis for distribution shift. Together with the spatial holdout, the future holdout supports a bounded claim: the evidence is within-farm spatial and temporal stress validation of routing semantics, not a value-prediction guarantee and not evidence that the rule carries to a different wind farm without local testing. The holdout summaries are retained as source-trace evidence in the final manifest rather than re-input from the older goal-table directory.

We add a positive deployment-drill result at exactly the level supported by the evidence. The drill is not Kelmarsh/Penmanshiel transfer and it does not undo the external failure reported below. It asks whether a deployable workflow can be executed inside WTB under stricter separation: use a small calibration

Table. Quasi-external deployment drill inside WTB: calibration-only boundary/gate-map selection followed by held-out testing.

Scenario	Calib. days	Runs	Min boundary cells	Held-out NMI/ARI	Bal. acc.	Decision	Boundary reading
future-period WTB proxy	2.0	5/5	67	0.8397/0.8907	0.9770	passes pre-specified held-out routing gate	supports calibration-to-held-out workflow inside WTB; not external-farm portability
east-turbine WTB proxy	7.0	5/5	103	0.8272/0.8781	0.9709	passes pre-specified held-out routing gate	supports calibration-to-held-out workflow inside WTB; not external-farm portability

window to select rated wind, pitch threshold, boundary support, and the gate-to-regime map; freeze those choices; then evaluate held-out routing once. In the future-period proxy, a 2-day calibration window followed by the future holdout passes in all 5/5 seeds, with mean held-out NMI/ARI 0.8397/0.8907 and balanced accuracy 0.9770. In the east-turbine proxy, a 7-day calibration window on held-out turbines also passes in all 5/5 seeds, with mean held-out NMI/ARI 0.8272/0.8781 and balanced accuracy 0.9709. The 2-day and 7-day windows are WTB proxy settings, not universal minimum durations for a new farm. The result proves an operational procedure inside the WTB envelope: local calibration window $\rightarrow$ boundary re-estimation $\rightarrow$ frozen held-out routing test. It does not prove that the WTB rule migrates to a new farm.

This drill also defines the operator-facing usability checks. Threshold choice is restricted to a pre-specified local grid and must be frozen before the held-out test. Calibration support is checked by the number of valid boundary cells, not only by elapsed days. The effective minimum calibration window is therefore governed by boundary-cell support, pitch observability or a validated pitch proxy, and a frozen held-out routing criterion. If pitch channels are missing, the site may use a documented pitch proxy only after showing overlap with wind speed, active power, and availability masks; otherwise the gate cannot be given a physical-router interpretation. If rated wind differs across turbine models, the rated-wind grid is estimated locally rather than copied from WTB. If SCADA gaps or availability masks remove too many boundary cells, the output is a data-coverage failure, not a weak positive site result. Finally, reserve use requires a separate high-ramp slice check because the gate-bin rule can reduce total cost by carrying too little reserve in ramp-heavy windows.

5.6 Ablation

The WTB ablation exposes a controllable accuracy-semantics frontier. The completed strict-mask 50-run ablation covers dense, unconstrained, balance-only, align-only, balance-align, boundary-forced, s-smoothed boundary-forced, context-aligned, anchor-only, and full variants across seeds 201–205, with the routing-control mechanism check passed for the three routing variants. The reading is deliberately paired: every ablation row must report RMSE together with NMI/ARI. With load balancing, regime alignment, and boundary forcing active, the selected corrected router reaches NMI/ARI 0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371. Boundary forcing remains an interpretability intervention with a measured forecasting price rather than a free accuracy improvement.

The ablation claim is therefore the shape of the frontier, not a ranking headline. The current strict-mask artifact records 50/50 expected runs complete and 15/15 routing-gate checked runs passed; the paper uses that record to keep the ablation as evidence for the RMSE/NMI tradeoff.

Figure 6 visualizes this frontier. Unconstrained MoE sits in the low-semantics region. L_{align} moves the gate sharply upward with a modest RMSE change. The force/smooth variants continue toward cleaner partitions while moving rightward on RMSE. The full physics-aligned MoE is the routed forecasting compromise; the boundary-forced variants occupy the semantic-repair end of the frontier.

5.7 Robustness and fairness checks

The stress and fairness checks rule out two easy explanations for the WTB result. The in-family comparisons use multiple strict-mask seeds, and the routed in-family variants share the same parameter count, making a pure capacity explanation unlikely. The completed baseline refresh sharpens the forecasting check: Graph WaveNet is lower than the boundary-forced router by about 10.39 RMSE on average, PatchTST is lower by about 8.06 RMSE, and Graph Transformer and GAT-GRU are close in error but

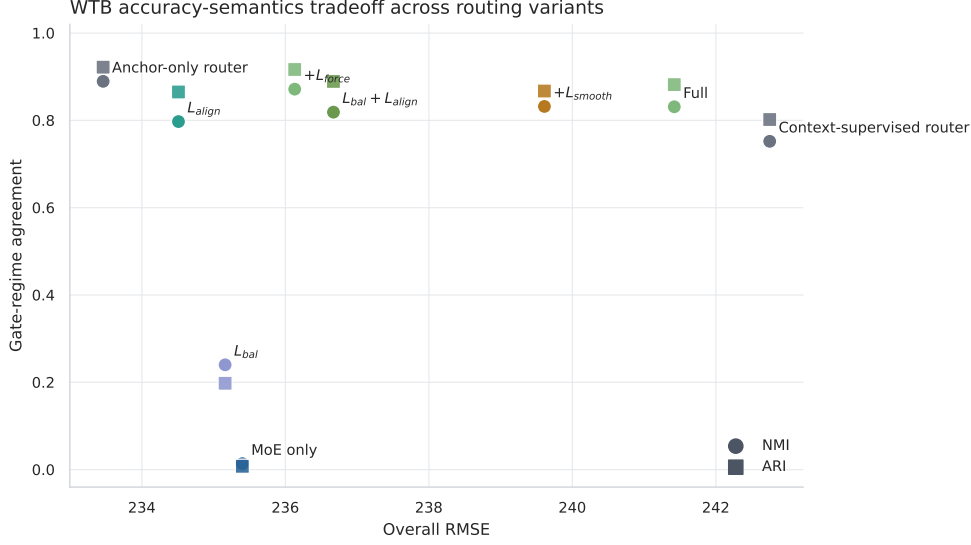


Figure 8: WTB accuracy-semantics tradeoff across routing variants. Circles show NMI and squares show ARI against overall RMSE. The frontier clarifies that stronger routing correction improves gate-regime agreement but can move the model away from the lowest forecast error.

Table. Parameters, training cost, and inference speed on strict-mask WTB runs.

Model	Params	Train h	Eval s	Win/s	Reading
PatchTST	111512	0.09	4.08	1221.7	fastest inference
Physics-Aligned MoE	110012	0.62	13.24	376.2	router adds interpretable responsibility at moderate cost
Boundary-forced router	110012	0.59	13.89	371.0	router adds interpretable responsibility at moderate cost
Graph WaveNet	123800	1.50	23.01	216.5	lowest strict-mask RMSE, slower

do not change the routing-accountability result. The evidence supports the same qualitative reading as the mechanism tables: the method earns its value through gate-regime responsibility, not headline error dominance. The corresponding ERA5 RMSE difference between physics-aligned MoE and the dense encoder is small and uncertain, about 0.009 (-0.061 to 0.085), and the expanded ERA5 baseline panel shows that even the lowest-RMSE learned baseline still trails persistence. The compute-cost table adds one more constraint: PatchTST is fastest, Graph WaveNet is slower but has the lowest strict-mask RMSE, and the routed models sit between them with about 110k trainable parameters and 13–14 s evaluation time for the strict test windows. These checks therefore support the routing-semantics claim and keep the accuracy and operating-scope claims bounded.

The strict replay, intervention, placebo, and time-forward tables below are the current stress evidence for the corrected-router claim.

We also add negative-control and intervention audits for the strict-mask WTB gate. The five-seed strict-mask set compares the actual operating labels with labels that should not preserve the MPPT-to-pitch boundary: an 18-step temporal shift, a one-day temporal shift, node permutation, within-time node shuffling, and global label shuffling. The boundary-forced router keeps high agreement with the actual labels (NMI/ARI 0.8716 +/- 0.0374 / 0.9166 +/- 0.0332 in the placebo summary). Actual labels exceed global shuffling by delta NMI/ARI 0.8716 / 0.9166, one-day temporal shift by 0.8324 / 0.8656, and node permutation by 0.3289 / 0.2825; all bootstrap intervals are positive. The per-seed placebo table shows that this separation is not carried by one lucky run: all five seeds retain large actual-vs-global and actual-vs-one-day-shift gaps. Spatial placebos remain partially aligned because turbines share fleet-level operating distributions, but the actual labels are still clearly separated.

The mechanism intervention gives the stronger test. Reloaded strict-mask checkpoints match their

Table 3. Replay-safe strict-mask WTB mechanism interventions.

Intervention	Delta RMSE	Delta switch RMSE	Drop NMI	Drop ARI
anchor_boundary_zero	2.6792 [0.7347, 4.5598]	2.9470 [0.7985, 5.1213]	0.7017 [0.6451, 0.7911]	0.8593 [0.8208, 0.9075]
anchor_wake_zero	0.0031 [-0.0016, 0.0110]	0.0032 [-0.0011, 0.0106]	0.0010 [0.0001, 0.0021]	0.0010 [0.0002, 0.0023]
anchor_all_zero	7.2037 [2.1205, 12.2869]	7.7592 [2.2208, 13.2587]	0.7191 [0.6579, 0.8230]	0.8638 [0.8244, 0.9179]

Table 4. Strict-mask WTB routing placebo separations.

Control	Delta NMI	Delta ARI
temporal_shift_18	0.5875 [0.5551, 0.6210]	0.5759 [0.5466, 0.6043]
temporal_shift_144	0.8324 [0.8020, 0.8692]	0.8656 [0.8345, 0.8954]
node_permutation	0.3289 [0.3016, 0.3659]	0.2825 [0.2581, 0.3100]
within_time_shuffle	0.3337 [0.3040, 0.3702]	0.2872 [0.2605, 0.3153]
global_shuffle	0.8716 [0.8411, 0.9086]	0.9166 [0.8879, 0.9422]

saved metrics for all five runs before intervention. Zeroing the boundary anchor raises overall RMSE by 2.6792 (95% bootstrap interval 0.7347 to 4.5598), raises switch-window RMSE by 2.9470 (0.7985 to 5.1213), and drops NMI/ARI by 0.7017 / 0.8593. The per-seed intervention table sharpens the interpretation. Boundary-anchor removal collapses NMI/ARI in every seed, but the RMSE penalty is heterogeneous: seeds 202 and 205 show near-zero or slightly negative RMSE deltas while still losing most of their routing agreement. The mechanism claim is therefore about gate responsibility, not an assertion that removing the anchor always worsens headline error. Zeroing the wake score is a near-null control: overall RMSE changes by 0.0031 (-0.0016 to 0.0110), and NMI/ARI drop by about 0.0010 / 0.0010. This moves the WTB claim beyond correlation or visualization: the gate depends on the intended boundary anchor, not on any arbitrary physical channel.

The same evidence stack is traceable to the exported source-data files in `artifacts/final_evidence_package/export`, including per-seed CSV files, protocol records, and SHA256 manifests.

5.8 Time-forward stress test

A post-hoc time-forward test-slice audit asks whether the strict-mask evidence survives later test windows. The answer is split. Routing semantics remain stable: in the late test block, the corrected router keeps NMI/ARI 0.8661 +/- 0.0393 / 0.9079 +/- 0.0364, and late-minus-early NMI is +0.0068 with a 95% bootstrap interval from 0.0021 to 0.0127. Forecasting accuracy, however, worsens in the late block. Late-minus-early overall RMSE is +50.3454 (46.8137 to 54.1758), and switch-window RMSE is +63.3549 (60.7291 to 65.3319). The added distribution-shift diagnostics explain this as a boundary condition: the late block has higher mean target power (+52.70), a large regime-share shift (total variation 0.2983, Jensen-Shannon divergence 0.0539), lower mask-valid coverage (-0.1446), and a lower mean wind-speed anchor (-0.3028) but much larger pitch action (+13.9845 in mean Pab). The gate still tracks the physical partition, but the value map faces a different operating distribution. This audit therefore supports a routing-semantics claim while weakening any time-stable forecasting claim. It is a stress test on saved test predictions, not a replacement for an external dataset or a pre-specified future-period holdout.

5.9 Sensitivity checks

Table 8 tests whether WTB routing recovery depends on a single hand-picked boundary or one large loss weight. Across three-seed sweeps, changing λ_{align} from 2500 to 7500 keeps NMI in the 0.8255–0.8546 range, and changing λ_{force} from 5000 to 15000 keeps NMI in the 0.8091–0.8406 range. The label-validity audit then re-labels the saved strict-mask gate outputs over the local grid rated-wind {10.0, 10.5, 11.0}

Table 5. Strict-mask WTB late-period distribution-shift diagnostics.

Model	Delta RMSE	Delta abs ramp	Regime TV	Regime JSD	Delta NMI
MoE + L_bal + L_align + L_force	50.3454 [46.8137, 54.1758]	-1104.6378 [-1104.6378, -1104.6378]	0.2983	0.0539	0.0068 [0.0021, 0.0127]

Table. WTB threshold/label validity audit for saved strict-mask gate outputs.

Rated wind	Pitch threshold	Runs	NMI		ARI	Shared valid
10.0	1.5	5	0.8724 ± 0.0404	0.9165 ± 0.0368		0.9956
10.0	2.0	5	0.8721 ± 0.0407	0.9164 ± 0.0369		0.9991
10.0	2.5	5	0.8717 ± 0.0410	0.9163 ± 0.0370		0.9990
10.5	1.5	5	0.8723 ± 0.0418	0.9165 ± 0.0372		0.9960
10.5	2.0	5	0.8710 ± 0.0420	0.9161 ± 0.0374		1.0000
10.5	2.5	5	0.8696 ± 0.0422	0.9157 ± 0.0374		0.9998
11.0	1.5	5	0.8687 ± 0.0426	0.9156 ± 0.0375		0.9909
11.0	2.0	5	0.8672 ± 0.0429	0.9151 ± 0.0377		0.9949
11.0	2.5	5	0.8655 ± 0.0429	0.9146 ± 0.0377		0.9949

and pitch-threshold $\{1.5, 2.0, 2.5\}$ without retraining. The worst grid point still has NMI/ARI $0.8655 \pm 0.0429 / 0.9146 \pm 0.0377$, and the minimum shared-valid support is 0.9909. This is a local rated-wind/pitch robustness readout, not a system-wide threshold sweep: u_{idle} , wake-score thresholds, turbine-specific control curves, and wider rated-wind/pitch grids remain untested. These rows are threshold/label validity checks and mask controls, not the semantic negative control. The semantic negative-control claim is reserved for `boundary_negative_controls_wtb` and the routing placebo audit, where deliberately incorrect or shifted labels are checked against the same intervention evidence. The main WTB result keeps the pre-specified operating-boundary rule.

The external Kelmarsh/Penmanshiel result is a failed boundary-condition test, and this negative result is one of the contributions of the paper. The source audit completes 80/80 runs, including both leave-one-farm-out directions and chronological checks, and it asks whether the default WTB operating boundary can be read on external farms. It cannot: the default WTB boundary gives mean routing NMI 0.4877, below the pre-specified 0.50 external-site routing criterion. The external farms show that an operating-boundary rule cannot be reused before local observability and boundary support are checked. In deployment terms, this is a fail-fast safety gate rather than a hidden generalization claim: if the local boundary cannot be recovered on held-out data, the router must be treated as a diagnostic warning and not as a reserve-control signal. We add validation-only local recalibration only to explain that failure mode. For each routing run, the validation gate output selects a farm-specific rated-wind/pitch-threshold pair from the pre-specified grid, and that selected boundary is evaluated once on the test gate output. This does not retrain the forecaster and does not rescue an out-of-site mechanism claim. Overall default test NMI rises only from 0.1324 to 0.1491 after recalibration, a recovery of +0.0167. Kelmarsh chronological runs recover more agreement, but Penmanshiel chronological and the leave-one-farm-out settings remain weak, so the failure is not merely a shifted threshold. The postmortem points to concrete boundary conditions rather than an unexplained error: pitch/proxy coverage differs from WTB, effective boundary cells can be sparse, control-state shares are imbalanced, farm scale and layout are smaller, turbine types and power curves differ, and the available sensor fields do not recreate the WTB anchor set. The practical message is protective: the boundary diagnostic prevents misdeployment when sensor fields, turbine geometry, or regime support do not match the WTB operating envelope.

The small calibration-window adaptation makes that boundary sharper. This 40/40 diagnostic is a second external test, separate from the WTB proxy drill: using only each external run’s validation gate outputs, it selects a local rated-wind/pitch pair and a majority-vote gate-to-regime map, then freezes both before evaluating held-out test gates. Chronological adapted balanced accuracy averages 0.4787, below the 0.50 diagnostic threshold. Penmanshiel chronological improves from 0.4000 to about 0.5000 balanced accuracy, but its adapted NMI remains 0.0000; Kelmarsh chronological declines, and the leave-one-farm-out Penmanshiel-to-Kelmarsh setting collapses. Therefore none of the external farms clears the

Table. Why external wind-farm boundary transfer failed and how the result is used.

Evidence gate	Observed	Decision	Bounded claim
pre-specified external routing criterion	80/80 runs complete; mean NMI 0.4877 below threshold 0.50; mean AR1 0.5112	failed external-site routing criterion	negative boundary-condition evidence only
validation-selected local boundary recalibration	default test NMI 0.1324 -> recalibrated 0.1491 (delta +0.0167)	small average recovery, not portability	local boundary re-estimation is required
small calibration-window gate-map adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	completed negative adaptation diagnostic	calibration window is insufficient without held-out pass
sensor and boundary support	Penmanshiel-to-Kelmarsh leave-one pitch-feature coverage 0.0000 and effective boundary cells 0	label not observable in this transfer direction	external physical-router interpretation is not supported
chronological-only local recalibration	boundary router chronological NMI recovery +0.0311; held-out test NMI after recalibration 0.2372	partial local recovery but still weak	generalization requires local held-out checks
single-site apparent improvement	Penmanshiel chronological adapted balanced accuracy 0.5000, but NMI remains 0.0000	not enough for a positive external claim	future work must satisfy a held-out criterion before site-level use

positive held-out routing criterion after local calibration. The result is a completed negative adaptation diagnostic: local boundary estimation is necessary, but this small window is not sufficient to recover usable routing at a new site.

The failure conditions are operationally informative. Kelmarsh has 6 Senvion MM92 turbines, whereas Penmanshiel has 14 retained Senvion MM82 turbines. Pitch-feature coverage and boundary support differ sharply, especially in leave-one-farm-out caches, and the regime shares move from WTB’s 39.47% idle, 52.00% MPPT, and 8.53% pitch-control partition to Kelmarsh’s MPPT-heavy chronological partition and Penmanshiel’s larger pitch-control share. These differences in sensor fields, pitch observability, turbine geometry, farm scale, power-curve distribution, and label balance are exactly the conditions under which the WTB operating-boundary rule must be re-estimated and tested before any site-level claim.

5.10 Local evidence protocol for a new wind farm

The external failure leads to a local evidence protocol rather than a site-transfer claim. A new wind farm must pass three evidence gates before the router can be interpreted physically. First, the site must pass a pitch-observability gate: wind speed, active power, turbine status or availability masks, and pitch-related channels must be present with enough valid overlap to construct MPPT and pitch-control labels. The minimum calibration window is not a transferable number of days. It is the smallest pre-declared window that contains enough effective boundary cells, pitch observability or a validated pitch proxy, and a frozen held-out routing check; if pitch is sparse, the window must be extended until those conditions are met. If mean pitch, blade-pitch channels, or a reliable pitch proxy are absent in the intended operating window, the boundary label is not identifiable and the router remains a statistical diagnostic.

Second, the site must pass a local-boundary-support gate. Rated wind speed, pitch threshold, boundary band, and gate-to-regime map must be re-estimated on a local calibration period, then frozen before held-out testing; selecting them on the test period would convert the protocol into a retrospective explanation. The calibration period must contain enough boundary cells, not only idle or MPPT-heavy operation. If local support is too sparse, the result may still be reported as a data-coverage diagnostic, but the WTB-derived boundary rule must not be reused as a physical explanation.

Third, the site must pass a held-out routing criterion before the gate is interpreted as a physical boundary signal. The frozen gate-to-regime map, rated-wind threshold, pitch threshold, and boundary band must clear held-out NMI ≥ 0.50 and balanced accuracy ≥ 0.50 , or a stricter site-specific criterion declared before test evaluation. The local idle/MPPT/pitch shares should also be compared with the WTB operating envelope, and transition-window shortage, violation rate, reserve energy, and high-ramp slices must be recomputed before any gate-bin reserve rule is used. A new site passes the protocol only if the held-out gate clears the pre-specified routing criterion and the reserve audit improves shortage and violations in the intended transition window at an acceptable reserve-energy cost. If any gate fails, the result remains a boundary-condition diagnosis; physical routing interpretation, new-farm operating evidence, and gate-bin reserve use are not supported.

6 Discussion

The main result is a rule for using physics-aligned routing in wind-power operations: correction strength should scale with regime observability and with the operating decision that will consume the gate. ERA5 already exposes the regime marker through sensible heat flux and its temporal variation, and persistence is

New wind-farm local evidence protocol: physical interpretation requires every check to pass

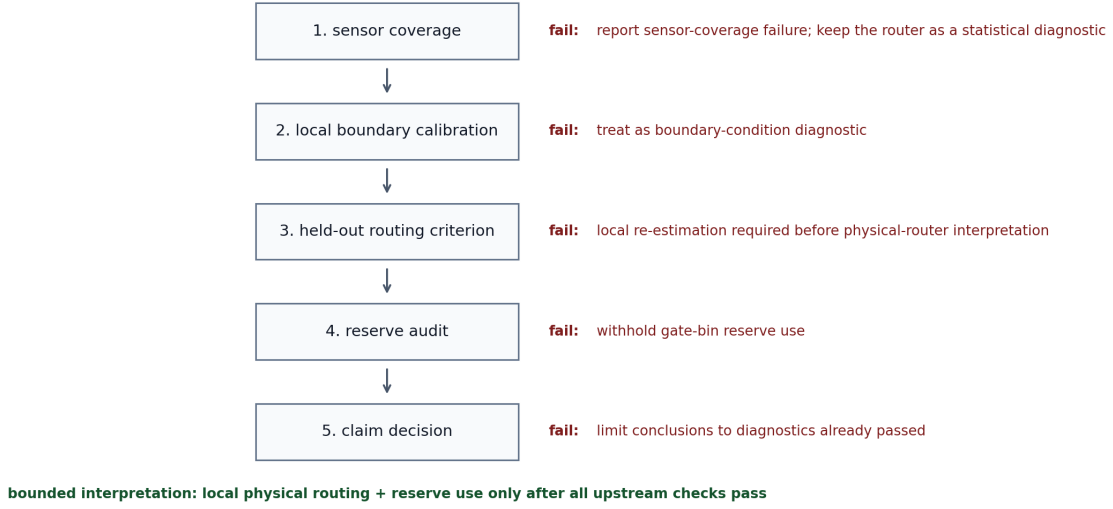


Figure 9: New wind-farm evidence protocol. The protocol moves from sensor coverage to local boundary calibration, held-out routing, reserve audit, and finally bounded operating claims.

hard to beat on RMSE. In that setting, the corrected gate improves regime alignment over Unconstrained MoE without creating a lower-error headline map. WTB is different. Wake interaction and blade-pitch control blur the MPPT-to-pitch boundary, so the router needs stronger repair to recover a useful partition. The WTB-ERA5 contrast is the paper’s main scientific insight: the same routing prior has different value depending on how much of the physical boundary is already visible in the observations.

The price is explicit. Graph WaveNet is the most stable black-box WTB forecaster, while LightGBM and XGBoost lag-feature baselines show that simple engineering predictors remain highly competitive. Inside the routed family, the full physics-aligned MoE is the forecasting compromise, while the boundary-forced router is the semantic-repair variant. Regime-anchor alignment buys a large share of the partition recovery with a modest RMSE change. Boundary forcing buys a cleaner gate at a clear forecasting cost. That is the paper’s practical claim: gate correction is a modeling choice with measurable error and reserve-policy tradeoffs.

For mechanism-aware forecasting studies, physics has to constrain the routing decision itself. Many physics-guided models act on outputs or latent states, which suits tasks where the goal is physical consistency after prediction. Here the decision is earlier: which local predictor takes responsibility near a mechanism change. The MPPT-to-pitch boundary is a physically meaningful stress point because it changes reserve shortage risk and ramp-risk interpretation even when fleet-level average error moves only modestly. Boundary-focused forcing encodes that stress point at the router. It sacrifices some RMSE to keep the gate from drifting into a semantically useless partition.

The method is appropriate when the operating boundary is part of the scientific question. Examples include turbine-control transitions, curtailment regimes, and geophysical stability states where the active mechanism matters, not only the scalar error. It is a poor choice for a pure leaderboard objective, for settings where persistence or a black-box graph model is already adequate, or for problems without a defensible regime anchor. In those cases, the extra routing losses add complexity without a clear mechanism-evidence return. The target use case is the middle ground: mechanism boundaries are important enough that interpretable responsibility can justify a measured error penalty. This positioning also explains why the strongest claim is about controlled responsibility assignment rather than broad cross-dataset superiority.

7 Limitations

The WTB correction depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those same channels also appear in the gate anchor, so the method deliberately constrains the router to a declared operating boundary. The reported threshold audit re-labels saved gate outputs over a local rated-wind/pitch grid and the lambda sweep varies the two dominant routing weights, so the main result is not a single-point artifact. The coverage is still partial: it does not exhaust all anchor thresholds, loss weights, wake-score cutoffs, wake-cone parameters, or turbine-control settings. It also lacks retrained no-Pab_mean, no-Wspd, lagged-Pab_mean, and lagged-Wspd ablations. The method is boundary-aware regularized routing, not a fully learned operating-state discovery system, and it cannot separate observable-anchor responsibility from structure learned through other SCADA channels.

The active-power anchor has a strict timing requirement. Patv is allowed only as the historical/anchor-time active-power measurement available when the forecast is issued; future active power remains the supervised target. This is a defensible SCADA forecasting convention, but it is also a portability constraint and a possible source of interpretive circularity. The current intervention evidence zeroes the boundary-anchor input and shows that routing agreement collapses, but it is not a retrained no-Patv or lagged-Patv ablation. Taken together with the missing wind-speed and pitch ablations above, the present evidence cannot claim to identify which part of the gate is pitch/wind observability, active-power status, or structure learned after anchoring. If a deployment environment delays active-power telemetry, changes channel definitions, or evaluates a decision before the current active-power value is available, the Patv anchor must be removed or lagged and the leakage, intervention, and reserve diagnostics must be rerun.

The expert-regime language is tied to the implemented logit convention. The first primary-regime logits are fixed to the operating labels during supervised alignment, which prevents seed-wise permutation for those anchored meanings. That makes cross-seed MPPT/pitch statements interpretable, but only for the anchored logits and only under the declared mapping. Unassigned experts and wake auxiliary logits should not be overread as universal turbine states.

The comparison is bounded. Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, persistence, physical power curve, XGBoost/LightGBM lag features, and a DLinear-style LTSF readout provide completed strict-mask WTB forecasting baselines, and ERA5 is checked against persistence plus Graph WaveNet, STGCN, PatchTST, and TCN. The strict-anchor-mask rerun covers the selected WTB corrected router, replay, intervention, placebo, time-forward audits, a synchronized baseline refresh, a spatial node-held-out stress test, a future-period holdout, a quasi-external WTB deployment drill, and the transition-window reserve audit. Larger general time-series backbones, broader graph-transformer variants, trained quantile-regression baselines, and scenario/distributional forecasters are outside the present benchmark.

The reserve audit is deliberately narrower than power-system dispatch. The study does not implement unit commitment, market clearing, a full probabilistic prediction-to-reserve pipeline, real dispatch integration, or a closed-loop operator simulation. The audit is therefore a normalized proxy decision: fixed empirical quantile bins are calibrated on validation shortfall and tested once on held-out predictions, with costs reported in normalized reserve-energy units rather than money. It is useful for ranking boundary-window risk tradeoffs under declared cost ratios, but it is not a market-price, delivery-constrained, or security-constrained dispatch study.

External wind-farm diagnostics are complete but fail the external-site routing criterion. The Kelmarsh/Penmanshiel audit reaches 80/80 completed runs in both leave-one-farm-out directions, while the default WTB boundary remains below the declared threshold. The added local recalibration table records how much agreement is recovered when rated-wind/pitch boundaries are re-estimated on validation gates and then fixed on test gates. The quasi-external drill shows that the calibration-to-held-out workflow can pass inside WTB, but it does not change the external-farm conclusion. The external taxonomy points to farm scale, turbine geometry, pitch observability, sensor-field coverage, power-curve differences, and regime-label imbalance as the main boundary conditions. The evidence therefore supports within-WTB

spatial and temporal stress validation of routing semantics and operating-boundary reserve diagnostics, plus an external diagnostic of when those semantics need local boundary re-estimation.

The time-forward audit is not an external validation. It is a post-hoc split of saved WTB test predictions into contiguous anchor-time blocks. It shows that the strict-mask gate keeps its physical partition in the late block, but it also shows a large late-test RMSE increase. The failure case is therefore explicit: late-period distribution shift preserves routing agreement while degrading value prediction. This is an important boundary condition: the method currently supports routing semantics, not time-stable forecasting accuracy.

The WTB ablation is scoped. The strict-mask 50-run ablation record is complete, but each cited row must report RMSE together with NMI/ARI. The ablation supports the observed accuracy-semantics tradeoff; it does not convert the routed model into the lowest-error predictor.

Statistical reliability is uneven across claim types. The completed strict-mask WTB baseline refresh provides five seeds each for Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST, and the selected boundary-forced router has five strict-mask seeds, five future-holdout seeds, and five seeds for each spatial-holdout routing comparator. ERA5 has five seeds for the in-family models and three seeds for the learned baselines. The quasi-external WTB deployment drill has five future-period seeds and five east-turbine seeds, but it remains a proxy workflow test. The engineering baselines and reserve-window audit are deterministic or sampled strict-mask diagnostics rather than five-seed neural families. Semantic negative controls are taken from `boundary_negative_controls_wtb`, and the mask/threshold sensitivity controls are reported as sensitivity checks only. The external Kelmarsh/Penmanshiel audit is complete at 80/80 but remains negative for new-site routing; the recalibration and small-calibration-window adaptation diagnostics are therefore reported as boundary-condition analyses, not as new training results or external-site evidence. The strongest statistical claims are therefore the strict-mask replay-safe routing-recovery, mechanism-intervention, failure-case transparency, and within-WTB spatial/temporal stress claims.

The paper studies two observability settings. WTB and ERA5 are enough for the intended contrast between weakly and strongly expressed regime markers, with ERA5 serving as a positive/contrast control rather than as the main performance contribution. Applying the same interpretation to other industrial or geophysical systems will require re-estimating the operating boundary, checking sensor availability, and rerunning the routing gate before making a local mechanism claim.

8 Conclusion

This paper treats non-stationary wind-power modeling as a routing-accountability problem around operating transitions. The completed strict-mask WTB comparison gives a blunt accuracy result: strong graph, transformer, and lag-feature baselines achieve lower mean RMSE than the boundary-forced routed model. The strict-anchor-mask evidence then tests the selected router more severely: across five seeds it recovers the operating partition, passes checkpoint replay, fails under boundary-anchor removal, survives placebo controls, remains meaningful under spatial and future-period holdouts, and keeps routing semantics in late test windows. The quasi-external WTB drill adds one positive operational procedure: a small calibration window can re-estimate the local boundary and gate map, freeze them, and pass a held-out future or held-out east-turbine test. This is within-WTB spatial and temporal stress evidence plus workflow evidence, not evidence for a new wind farm. ERA5 carries the same lesson as an observability contrast: when the regime marker is visible and persistence is strong, physics-aligned routing improves gate-regime agreement more than headline forecasting accuracy.

The practical conclusion follows directly. When the regime marker is visible, routing regularization calibrates an already available partition. When the marker is partly hidden by control logic, physics guidance at the gate can repair a weakly identified partition and provide a transition-window reserve/ramp-risk diagnostic, but it trades against forecast error and does not create a full-sample reserve policy. Physical

forecasting with multiple local response laws therefore needs supervision over responsibility as well as supervision over values. The completed external wind diagnostic shows that this responsibility claim must be re-established when turbine geometry, pitch observability, sensor fields, power curves, or label distributions change; a small validation-window adaptation did not clear the held-out routing criterion in Kelmarsh/Penmanshiel. The gate is part of the model’s scientific claim, and the right question is whether that claim is worth its measured error, reserve, stress-test, and efficiency costs in the target operating setting. Until a new wind farm passes the local evidence protocol, the method should be described only as within-WTB operating-boundary auditability plus external failure-boundary diagnosis plus transition-window reserve diagnosis.

9 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT/Codex to support language editing, consistency checking, and submission-material drafting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

10 Code and data availability

The experiments use four public source families: the KDD Cup 2022 wind-farm SCADA benchmark, ERA5 reanalysis fields, and the Kelmarsh and Penmanshiel wind-farm SCADA records. Source datasets remain available from their original providers. The release package records provider links, checksums, and reconstruction scripts rather than redistributing any source files whose original terms require provider-side access. The Kelmarsh and Penmanshiel sources are tracked in the reproducibility manifest as CC-BY-4.0 source-data protocols; their completed 80-run audit, recalibration table, and small-calibration-window adaptation table are cited only as boundary-condition evidence because new-site reuse is not claimed. The final evidence export includes table, figure, source-data, protocol, and trace manifests; each manuscript table or figure is linked to source files, seed lists, run tables, raw prediction paths, checkpoint hashes, and SHA256 hashes where file size permits. The table, figure, and source-artifact indices in `artifacts/final_evidence_package/manifest/evidence_manifest_final.json` are the source/checksum authority for the manuscript artifacts, including the Applied Energy reserve diagnostics, bounded-claim table, quasi-external WTB deployment drill, and external-site diagnostics. The reproducibility manifest in `artifacts/reproducibility_manifest_final/reproducibility_manifest.json` reports 53/53 required files present, complete strict model weights, a recorded Python/CUDA environment, and no active experiment gaps. Code, configuration files, scripts, derived caches when licensing permits, model-weight/checkpoint hashes, seed-level CSV summaries, final figure source data, and evidence manifests can be released with the manuscript. Raw third-party data that cannot be redistributed will be accompanied by download instructions and rebuilding scripts. The one-command reproduction package rebuilds the strict evidence package, the engineering baseline panel, the transition-window reserve audit, the reserve protocol, the quasi-external WTB deployment drill, the external recalibration and small-calibration diagnostics, the final evidence manifest, and the reproducibility manifest with relative paths. The study uses turbine and atmospheric measurements only and involves no human participants or human-subject data.

Table A1. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

11 Appendix A. Training and implementation details

11.1 Training loop summary

The optimization procedure is identical across runs except for the routing penalties that are activated in each setting and the anchor labels that are available.

Algorithm A1 Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors R , auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

 Build node-level graph context from the precomputed graph tensors

 Encode X with the directed-diffusion GRU to obtain node context H

 Compute gate logits Z and soft routing probabilities G

 Compute expert forecasts and aggregate them with G

 Evaluate L_{pred}

 If MoE mode is active:

 add L_{bal} on top- K routing statistics

 add L_{align} on valid primary regime anchors

 if WTB: add L_{force} on MPPT/pitch samples

 if WTB: add L_{aux} on wake labels

 add L_{smooth} on the final graph snapshot

 Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

11.2 Shared constants

11.3 Dataset-specific thresholds and routing weights

Bai, L., Yao, L., Li, C., Wang, X., Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting, in: Advances in Neural Information Processing Systems. pp. 17804–17815.

Bremnes, J.B., 2004. Probabilistic wind power forecasts using local quantile regression. Wind Energy 7, 47–54. <https://doi.org/10.1002/we.107>

Cao, Y., Wang, J., 2026. ECTO: Exogenous-conditioned temporal operator for ultra-short-term wind

Table A2. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

Table A3. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

power forecasting.

- Daenens, S., Verstraeten, T., Daems, P.-J., Nowé, A., Helsen, J., 2025. Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data. *Wind Energy Science* 10, 1137–1152. <https://doi.org/10.5194/wes-10-1137-2025>
- Doherty, R., O’Malley, M., 2005. A new approach to quantify reserve demand in systems with significant installed wind capacity. *IEEE Transactions on Power Systems* 20, 587–595. <https://doi.org/10.1109/TPWRS.2005.846206>
- Ela, E., Milligan, M., Kirby, B., 2011. Operating reserves and variable generation (No. NREL/TP-5500-51978). National Renewable Energy Laboratory, Golden, CO. <https://doi.org/10.2172/1023095>
- Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 1–39.
- Gallego-Castillo, C., Cuerva-Tejero, A., Lopez-Garcia, O., 2015. A review on the recent history of wind power ramp forecasting. *Renewable and Sustainable Energy Reviews* 52, 1148–1157. <https://doi.org/10.1016/j.rser.2015.07.154>
- Guo, S., Lin, Y., Feng, N., Song, C., Wan, H., 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting, in: *Proceedings of the AAAI Conference on Artificial Intelligence*. pp. 922–929. <https://doi.org/10.1609/aaai.v33i01.3301922>
- Haq, I.U., Kumar, A., Rathore, P.S., 2025. Machine learning approaches for wind power forecasting: A comprehensive review. *Discover Applied Sciences* 7, 1139. <https://doi.org/10.1007/s42452-025-07675-x>
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J.-R., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., Rosnay, P. de, Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.-N., 2020. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999–2049. <https://doi.org/10.1002/qj.3803>
- Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E., 1991. Adaptive mixtures of local experts. *Neural Computation* 3, 79–87. <https://doi.org/10.1162/neco.1991.3.1.79>
- Jordan, M.I., Jacobs, R.A., 1994. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation* 6, 181–214. <https://doi.org/10.1162/neco.1994.6.2.181>

- Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3, 422–440. <https://doi.org/10.1038/s42254-021-00314-5>
- Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., Shekhar, S., Samatova, N.F., Kumar, V., 2017. Theory-guided data science: A new paradigm for scientific discovery from data. *IEEE Transactions on Knowledge and Data Engineering* 29, 2318–2331. <https://doi.org/10.1109/TKDE.2017.2720168>
- Khodayar, M., Wang, J., 2019. Spatio-temporal graph deep neural network for short-term wind speed forecasting. *IEEE Transactions on Sustainable Energy* 10, 670–681. <https://doi.org/10.1109/TSTE.2018.2844102>
- Li, S., Li, X., Jiang, Y., Yang, Q., Lin, M., Peng, L., Yu, J., 2025. A novel frequency-domain physics-informed neural network for accurate prediction of 3D spatio-temporal wind fields in wind turbine applications. *Applied Energy* 386, 125526. <https://doi.org/10.1016/j.apenergy.2025.125526>
- Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: *International Conference on Learning Representations*.
- Nielsen, H.A., Madsen, H., Nielsen, T.S., 2006. Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts. *Wind Energy* 9, 95–108. <https://doi.org/10.1002/we.180>
- Park, J., Park, J., 2019. Physics-induced graph neural network: An application to wind-farm power estimation. *Energy* 187, 115883. <https://doi.org/10.1016/j.energy.2019.115883>
- Parsa, S.M., 2025. Physics-informed machine learning meets renewable energy systems: A review of advances, challenges, guidelines, and future outlooks. *Applied Energy* 402, 126925. <https://doi.org/10.1016/j.apenergy.2025.126925>
- Pinson, P., 2013. Wind energy: Forecasting challenges for its operational management. *Statistical Science* 28, 564–585. <https://doi.org/10.1214/13-STS445>
- Pinson, P., Chevallier, C., Kariniotakis, G.N., 2007. Trading wind generation from short-term probabilistic forecasts of wind power. *IEEE Transactions on Power Systems* 22, 1148–1156. <https://doi.org/10.1109/TPWRS.2007.901117>
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* 378, 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>
- Read, J.S., Jia, X., Willard, J., Appling, A.P., Zwart, J.A., Oliver, S.K., Karpatne, A., Hansen, G.J.A., Hanson, P.C., Watkins, W., Steinbach, M., Kumar, V., 2019. Process-guided deep learning predictions of lake water temperature. *Water Resources Research* 55, 9173–9190. <https://doi.org/10.1029/2019WR024922>
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: *International Conference on Learning Representations*.
- Shi, S., Han, L., Liu, Y., others, 2025. Time-MoE: Billion-scale time series foundation models with mixture of experts, in: *International Conference on Learning Representations*.
- Tautz-Weinert, J., Watson, S.J., 2017. Using SCADA data for wind turbine condition monitoring: A review. *IET Renewable Power Generation* 11, 382–394. <https://doi.org/10.1049/iet-rpg.2016.0248>
- Tian, J., Ding, Y., Xu, R., Miao, H., Guo, C., Yang, B., 2026. ARROW: An adaptive rollout and routing method for global weather forecasting.
- Wang, J., Botterud, A., Bessa, R., Keko, H., Carvalho, L., Issicaba, D., Sumaili, J., Miranda, V., 2011. Wind power forecasting uncertainty and unit commitment. *Applied Energy* 88, 4014–4023. <https://doi.org/10.1016/j.apenergy.2011.04.011>
- Wu, Z., Pan, S., Long, G., Jiang, J., Chang, X., Zhang, C., 2020. Connecting the dots: Multivariate time series forecasting with graph neural networks, in: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 753–763. <https://doi.org/10.1145/3394486.3403118>

- Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C., 2019. Graph WaveNet for deep spatial-temporal graph modeling, in: Proceedings of the 28th International Joint Conference on Artificial Intelligence. pp. 1907–1913. <https://doi.org/10.24963/ijcai.2019/264>
- Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: Proceedings of the 27th International Joint Conference on Artificial Intelligence. pp. 3634–3640. <https://doi.org/10.24963/ijcai.2018/505>
- Yu, M., Zhang, Z., Li, X., Yu, J., Gao, J., Liu, Z., You, B., Zheng, X., Yu, R., 2020. Superposition graph neural network for offshore wind power prediction. *Future Generation Computer Systems* 113, 145–157. <https://doi.org/10.1016/j.future.2020.06.024>
- Zehtabiyani-Rezaie, N., Iosifidis, A., Abkar, M., 2023. Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability. *PRX Energy* 2, 013009. <https://doi.org/10.1103/PRXEnergy.2.013009>
- Zhang, J., Zhao, X., 2021. Spatiotemporal wind field prediction based on physics-informed deep learning and LIDAR measurements. *Applied Energy* 288, 116641. <https://doi.org/10.1016/j.apenergy.2021.116641>
- Zhang, Y., Wang, J., Wang, X., 2014. Review on probabilistic forecasting of wind power generation. *Renewable and Sustainable Energy Reviews* 32, 255–270. <https://doi.org/10.1016/j.rser.2014.01.033>
- Zhou, J., Lu, X., Xiao, Y., Tang, J., Su, J., Li, Y., Liu, J., Lyu, J., Ma, Y., Dou, D., 2024. SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array. *Scientific Data* 11, 649. <https://doi.org/10.1038/s41597-024-03427-5>
- Zhou, Z., Botterud, A., Wang, J., Bessa, R.J., Keko, H., Sumaili, J., Miranda, V., 2013. Application of probabilistic wind power forecasting in electricity markets. *Wind Energy* 16, 321–338. <https://doi.org/10.1002/we.1496>